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Supplementary Benchmark Results for fastEmbedR

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Abstract

This supplement records the completed observations and explicit failures of diagnostic campaign 20260930T203319Z: workflow timing and memory, embedding quality, neighbor recall, support width, CPU scaling, held-out transformation, landmarking, PCA, and graph clustering. A separate September 29 campaign supplies the illustrative embedding gallery. Neither campaign passed its final audit; results are not release-level speed claims.

Keywords: dimensionality reduction, t-SNE, UMAP, reproducible benchmarks.

1. Workflow elapsed times

Table S1 combines the R and direct-Python results for PCA, t-SNE, and UMAP without treating their timing boundaries as equivalent. It gives every observed completed time, marks single-fit measurements, and identifies failed or missing combinations. The source campaign used full data-set matrices and failed its final audit; the times are descriptive diagnostics, not publication-grade speed ratios. "R call" is the complete public fit; "Py fit" is the direct Python estimator fit, excluding process startup. An asterisk marks a completed single fit without timing repetitions. T denotes a time limit, M a confirmed memory limit, F another failure, K an unclassified kill, U unsupported, C cancelled, I a completed fit without usable time, and – not run or no status.

Table S1: Observed elapsed time in seconds from campaign 20260930T203319Z (diagnostic, failed audit). R call and direct Python fit use different timing boundaries. All fits use the full source matrix.

PCA, t-SNE, and UMAP methods

A. PCA, data sets 1/2

Method	Scope	COIL-20	USPS	Fashion	FlowRepo.	flow18	MNIST
fastEmbedR [CPU]	R call	2.89	0.26	7.64	19.9	0.63	7.52
irlba [CPU]	R call	0.89	0.099	1.78	455	0.75	1.77
fastEmbedR [CUDA]	R call	0.020	0.006	0.039	1.56	0.042	0.043
scikit-learn PCA [Python]	Py fit	0.45	0.050	0.96	19.0	1.10	1.03
cuML [CUDA]	Py fit	6.05	0.027	0.33	2.68	0.070	0.34

A. PCA, data sets 2/2

Method	Scope	ImageNet	MetRef	mass41	Tabula	Retina
fastEmbedR [CPU]	R call	230	0.035	0.72	0.81	0.38
irlba [CPU]	R call	101	0.026	38.7	0.61	0.46
fastEmbedR [CUDA]	R call	0.89	0.005	0.054	0.017	0.010
scikit-learn PCA [Python]	Py fit	16.0	0.006	0.68	0.31	0.086
cuML [CUDA]	Py fit	19.7	0.005	0.10	0.044	0.021

B. t-SNE, data sets 1/2

Method	Scope	COIL-20	USPS	Fashion	FlowRepo.	flow18	MNIST
Rtsne [CPU]	R call	12.0	15.4	426	T	1726*	825
FIt-SNE [CPU]	R call	15.0	6.64	84.1	3702*	736	109
fastEmbedR [CPU]	R call	18.8	11.9	125	3354*	300	129
fastEmbedR [CUDA]	R call	0.86	0.22	1.39	F	6.07	1.40
scikit-learn t-SNE [Python]	Py fit	5.19	66.8	711	T	T	774
openTSNE [Python]	Py fit	40.9	108	242	T	1098*	241
cuML [CUDA]	Py fit	5.94	0.41	2.42	F	51.4	2.39

B. t-SNE, data sets 2/2

Method	Scope	ImageNet	MetRef	mass41	Tabula	Retina
Rtsne [CPU]	R call	T	0.86	2746*	157	70.1
FIt-SNE [CPU]	R call	2541*	3.81	390	105	59.1
fastEmbedR [CPU]	R call	1672*	9.80	300	86.5	27.7
fastEmbedR [CUDA]	R call	505	0.16	6.29	3.24	0.42
scikit-learn t-SNE [Python]	Py fit	T	1.54	T	648	226
openTSNE [Python]	Py fit	1484*	11.8	584	125	97.0
cuML [CUDA]	Py fit	183	0.19	48.7	2.63	1.27

C. UMAP, data sets 1/2

Method	Scope	COIL-20	USPS	Fashion	FlowRepo.	flow18	MNIST
umap [CPU]	R call	181	145	2704*	T	T	2223*
uwot [CPU]	R call	14.6	7.12	62.8	6016*	1302*	65.9
uwot fast SGD [CPU]	R call	13.7	4.56	51.4	3958*	729	52.6
fastEmbedR fuzzy [CPU]	R call	6.77	3.29	79.7	2987*	319	86.2
fastEmbedR binary [CPU]	R call	6.99	7.60	127	5212*	762	113
fastEmbedR fuzzy [CUDA]	R call	0.77	0.12	1.03	F	1.14	1.03
fastEmbedR binary [CUDA]	R call	0.80	0.12	1.09	F	1.69	1.10
umap-learn [Python]	Py fit	11.6	8.36	85.5	T	1636*	52.1
cuML [CUDA]	Py fit	0.18	0.14	0.87	F	9.11	0.87

Runtime table, continued

C. UMAP, data sets 2/2

Method	Scope	ImageNet	MetRef	mass41	Tabula	Retina
umap [CPU]	R call	T	4.13	T	1806*	469
uwot [CPU]	R call	2643*	1.36	590	106	25.5
uwot fast SGD [CPU]	R call	1710*	0.80	485	61.9	13.5
fastEmbedR fuzzy [CPU]	R call	1559*	0.22	307	26.5	17.6
fastEmbedR binary [CPU]	R call	2320*	0.44	756	75.8	34.3
fastEmbedR fuzzy [CUDA]	R call	506	0.086	1.51	0.35	0.14
fastEmbedR binary [CUDA]	R call	497	0.088	2.10	0.44	0.19
umap-learn [Python]	Py fit	5654*	1.48	885	64.3	28.1
cuML [CUDA]	Py fit	307	0.18	6.51	0.72	0.30

2. Comparator grid and timing boundaries

In the revised two-dimensional t-SNE protocol, **fastEmbedR** reports its selected grid in each fit. The pinned FIt-SNE wrapper exposes interpolation points, minimum intervals, and intervals per coordinate unit. The benchmark requests three interpolation points and minimum interval counts of 43 below 40,000 rows and 86 otherwise. FIt-SNE rounds these to supported box counts of at least 50 and 90, respectively, corresponding to at least 150 and 270 interpolation nodes per axis. The realized grid can be larger if the current coordinate span requires it; the wrapper does not return that size. These settings approximate rather than exactly match the **fastEmbedR** CPU 128- and 256-cell policies. FIt-SNE three-dimensional Barnes–Hut runs do not use the interpolation grid.

The tested cuML 26.6.0 implementation fixes three interpolation points and 50 intervals per axis in its two-dimensional FFT source, yielding a 300-cell padded transform axis. Its public Python t-SNE constructor does not expose a grid-resolution parameter, so a matched-grid cuML comparison would require modifying and rebuilding cuML. The benchmark records this fixed source policy, but does not mislabel the runs as grid matched. UMAP methods do not use this t-SNE FFT grid.

The September 30 workflow campaign has five same-seed timing repetitions after warm-up for timing-eligible methods; jobs reaching the long-run budget may have one completed fit. All input matrices were fitted in full. The 7,200-second per-method limit and failed campaign audit are reflected in Table S1. The embedding gallery below comes from a separate, capped September 29 campaign and must not be read as visual evidence for the September 30 timings.

3. Data-set-level CUDA embedding quality

Table S2 reports trustworthiness, Preserve@30, and sampled t-SNE KL for every CUDA **fastEmbedR** and cuML pair. The methods use the same fitted row count and evaluation sample within a data set, but differ in affinity support and timing boundary. KL values across those objectives are not directly comparable.

Table S2: CUDA embedding quality in diagnostic campaign 20260930T203319Z. Entries are fastEmbedR / cuML on fixed sampled rows; n is the fitted matrix size. Status is fastEmbedR / cuML; – marks unavailable metrics. Different affinity supports make sampled t-SNE KL descriptive, not a matched-objective test.

A. Compact-support t-SNE / cuML t-SNE

Data set	Fitted n	Trustworthiness	Preserve@30	Sampled KL	Status
COIL-20	1,440	0.939 / 0.934	0.627 / 0.623	1.480 / 1.317	success / success
USPS	11,000	0.952 / 0.940	0.468 / 0.462	2.024 / 2.006	success / success
Fashion	70,000	0.973 / 0.966	0.485 / 0.491	1.877 / 1.744	success / success
FlowRepo.	5,220,347	– / –	– / –	– / –	failed / failed
flow18	1,000,021	0.967 / 0.951	0.445 / 0.422	2.049 / 1.857	success / success
MNIST	70,000	0.941 / 0.931	0.437 / 0.443	2.316 / 2.164	success / success
ImageNet	1,281,167	0.667 / 0.635	0.167 / 0.124	3.510 / 3.310	success / success
MetRef	873	0.848 / 0.797	0.442 / 0.358	1.846 / 2.153	success / success
mass41	965,282	0.968 / 0.949	0.407 / 0.396	2.128 / 1.960	success / success
Tabula	100,102	0.962 / 0.940	0.632 / 0.592	1.736 / 1.596	success / success
Retina	44,808	0.912 / 0.900	0.432 / 0.428	1.970 / 1.916	success / success

B. Fuzzy UMAP / cuML UMAP

Data set	Fitted n	Trustworthiness	Preserve@30	Status
COIL-20	1,440	0.908 / 0.920	0.613 / 0.620	success / success
USPS	11,000	0.885 / 0.883	0.358 / 0.363	success / success
Fashion	70,000	0.962 / 0.967	0.463 / 0.463	success / success
FlowRepo.	5,220,347	0.960 / 0.967	0.445 / 0.443	success / success
flow18	1,000,021	0.962 / 0.961	0.435 / 0.432	success / success
MNIST	70,000	0.939 / 0.935	0.452 / 0.444	success / success
ImageNet	1,281,167	0.585 / 0.647	0.104 / 0.161	success / success
MetRef	873	0.836 / 0.821	0.428 / 0.411	success / success
mass41	965,282	0.968 / 0.967	0.404 / 0.407	success / success
Tabula	100,102	0.923 / 0.970	0.607 / 0.643	success / success
Retina	44,808	0.900 / 0.904	0.406 / 0.411	success / success

4. Backend quality and t-SNE support

Table S3 reports full-workflow quality for every completed CPU and CUDA data-set pair. The CPU FlowRepository full-workflow quality run was incomplete; matched-neighbor quality from a different boundary is not substituted. Trustworthiness and Preserve@30 use fixed quality-sample rows. The low ImageNet values reflect this feature matrix's limited two-dimensional neighborhood preservation; ImageNet is used primarily as a scalability stress test. Final KL was not retained for the CUDA full-workflow diagnostic and is shown as missing, not zero.

The $k = P$ affinity support makes the target perplexity the maximum possible entropy of each conditional distribution. At $P = 30$, normalized conditional entropy was approximately 1.000 for $k = P$ and 0.761 for $k = 3P$ in an exact-neighbor affinity diagnostic. Table S4 gives paired time and quality summaries for $P = 10$, 20, and 30. KL values across support widths are objectives with different affinities, not comparable optimization losses for one common objective.

5. Scaling, transformation, and landmarks

The strong-scaling experiment used identical inputs at 1, 2, 4, 8, and 12 CPU threads on COIL20, MNIST, flow18, and ImageNet. Times are complete workflows; stage-level values, including search, initialization, graph or affinity construction, and optimization, remain in the machine-readable campaign. The ratios in Table S5 are relative to each data set's one-thread result.

Held-out transformation fits a reference on training rows and projects genuinely unseen queries without moving the reference. Table S6 contrasts query-to-reference Preserve@30 with joint embedding of those queries. Times are transform calls, not complete fit-plus-transform workflows.

The 20-percent landmark experiment instead reconstructs rows omitted from the fitted subset. It is a different statistical task from held-out prediction. In Table S7, elapsed and host-RSS ratios are landmark/full; ratios below one favor landmarking. The GPU peak ratio is unavailable where device sampling was inadequate. In particular, the CUDA UMAP median runtime and host-RSS ratios exceed one. Procrustes agreement is especially modest for fuzzy UMAP.

6. Secondary analyses and memory

Binary UMAP discards fuzzy edge weights and therefore changes the embedding objective. It is retained as an adjacency sensitivity analysis. Table S8 reports its observed time and Preserve@30 change, without treating it as a faster fuzzy-UMAP setting.

Table S9 separates complete-worker host RSS from sampled device-wide CUDA memory above the pre-worker allocator baseline. Device sampling may miss short peaks; missing samples remain missing. These memory boundaries are not the direct-Python fit-only timing boundary.

CPU and CUDA PCA were compared with dense centered singular value decomposition on 11 fixed subsets. Signs and equivalent subspace rotations were aligned before score and loading comparisons. **irlba** is an approximate performance comparator, not the numerical reference. ImageNet CPU rank-2 score agreement was 0.783, an unresolved outlier for that configuration.

Table S3: File-to-layout quality from campaign 20260930T203319Z. Each entry is CPU / CUDA, over successful full-workflow seeds. A missing CPU FlowRepository result is shown as –. KL is the production t-SNE objective; UMAP has no KL entry. The campaign audit failed.

Data set	Trustworthiness	Preserve@30	Final KL
Compact-support t-SNE			
COIL-20	0.938 / 0.940	0.630 / 0.627	0.794 / –
Fashion-MNIST	0.972 / 0.973	0.483 / 0.484	2.697 / –
flow18	0.960 / 0.966	0.438 / 0.444	2.887 / –
ImageNet	0.646 / 0.658	0.155 / 0.158	2.287 / –
Retina	0.906 / 0.912	0.424 / 0.431	2.941 / –
mass41	0.963 / 0.968	0.401 / 0.408	2.905 / –
MetRef	0.848 / 0.847	0.444 / 0.442	1.850 / –
MNIST	0.939 / 0.941	0.436 / 0.437	2.903 / –
Tabula Muris	0.949 / 0.960	0.619 / 0.634	2.183 / –
USPS	0.949 / 0.951	0.467 / 0.467	2.167 / –
FlowRepository	– / 0.964	– / 0.440	– / –
Fuzzy UMAP			
COIL-20	0.915 / 0.908	0.612 / 0.612	– / –
Fashion-MNIST	0.971 / 0.961	0.480 / 0.465	– / –
flow18	0.960 / 0.961	0.443 / 0.436	– / –
ImageNet	0.582 / 0.589	0.105 / 0.107	– / –
Retina	0.905 / 0.902	0.416 / 0.408	– / –
mass41	0.971 / 0.968	0.413 / 0.408	– / –
MetRef	0.826 / 0.835	0.411 / 0.420	– / –
MNIST	0.940 / 0.938	0.456 / 0.451	– / –
Tabula Muris	0.934 / 0.922	0.613 / 0.599	– / –
USPS	0.889 / 0.882	0.358 / 0.354	– / –
FlowRepository	– / 0.961	– / 0.442	– / –

Table S4: Support-width sensitivity in campaign 20260930T203319Z. Each median uses the same data sets with valid timing and quality at both support widths. Times cover precomputed-neighbor t-SNE calls with 250 exaggerated and 500 normal iterations, not matrix workflows. KL values describe different affinity objectives and are not matched-objective ratios.

Backend	P	k/P	Data sets	Seconds	Trust	Preserve@30	KL
CPU	10	1	4	7.76	0.914	0.424	2.194
CPU	10	3	4	8.27	0.897	0.410	2.046
CUDA	10	1	4	0.05	0.920	0.431	2.161
CUDA	10	3	4	0.07	0.907	0.416	2.084
CPU	20	1	4	7.62	0.935	0.449	2.063
CPU	20	3	4	8.44	0.921	0.429	1.846
CUDA	20	1	4	0.06	0.935	0.449	2.038
CUDA	20	3	4	0.09	0.927	0.434	1.951
CPU	30	1	10	54.67	0.944	0.443	1.992
CPU	30	3	10	54.94	0.933	0.430	1.735
CUDA	30	1	11	0.29	0.952	0.442	1.987
CUDA	30	3	11	0.69	0.940	0.433	1.727

Table S5: CPU full-workflow strong scaling in campaign 20260930T203319Z. Speedups at 2–12 threads are relative to the one-thread elapsed time in the same row.

Data set	Method	1-core s	2-core	4-core	8-core	12-core
COIL20	UMAP	14.5	1.90	2.19	2.02	2.13
MNIST	UMAP	246.6	1.44	2.64	3.65	3.40
flow18	UMAP	632.8	1.10	1.87	2.03	2.37
imagenet	UMAP	4135.6	1.28	2.33	2.02	1.49
COIL20	t-SNE	19.0	1.88	2.16	1.95	2.05
MNIST	t-SNE	300.4	1.52	2.46	3.21	2.94
flow18	t-SNE	553.9	1.23	1.85	2.30	2.51
imagenet	t-SNE	4052.9	1.33	2.31	2.09	1.47

Table S6: Held-out transformation in campaign 20260930T203319Z. Each value is the median of data-set medians over successful seeds. Query/reference Preserve@30 is compared with joint embedding of the same queries; the reference is held fixed during transformation.

Method	Backend	Data sets	Transform s	Preserve@30	Joint	Label KNN	Ref. shift
t-SNE	CPU	11	64.684	0.371	0.393	0.784	0
UMAP	CPU	11	65.976	0.343	0.346	0.776	0
t-SNE	CUDA	11	0.621	0.374	0.392	0.786	0
UMAP	CUDA	11	0.543	0.351	0.340	0.780	0

Table S7: Landmark reconstruction over 11 data sets in campaign 20260930T203319Z. Each row is the median of data-set medians for a 20-percent landmark fraction. RSS covers isolated fit workers; – means insufficient device-memory evidence.

Method	Backend	Data sets	Time ratio	Preserve@30 Δ	Procrustes	RSS ratio	GPU ratio
t-SNE	CPU	11	0.41	-0.006	0.86	0.98	–
Fuzzy UMAP	CPU	11	0.34	-0.011	0.52	1.01	–
t-SNE	CUDA	11	0.77	-0.006	0.95	1.09	–
Fuzzy UMAP	CUDA	11	1.29	-0.006	0.47	1.07	–

Table S8: Binary-adjacency versus fuzzy UMAP in campaign 20260930T203319Z. Ratios and quality changes are diagnostic medians over completed data sets; binary UMAP changes the graph objective and is not a drop-in speed substitute.

Workflow	Data sets	Binary/fuzzy time	Preserve@30 change
r_cpu	9	1.986	0.003
r_cuda	11	1.053	0.008

Table S9: Observed peak memory in campaign 20260930T203319Z (diagnostic, failed audit). Host RSS covers the complete worker, including data loading and quality assessment; it is not fit-only memory. CUDA values are sampled device-memory increases above the pre-worker baseline (nominal 0.2-s polling plus query time), so short peaks may be underestimated. Values are GiB; – indicates no valid measurement. Counts show valid data sets for each memory metric; missing measurements are archived separately.

A. Host peak RSS

Method	Completed	Measured	Median	Maximum
fastEmbedR PCA [CPU]	11	11	1.66	36.90
irlba [CPU]	11	11	1.80	33.98
fastEmbedR PCA [CUDA]	11	11	1.85	37.08
scikit-learn PCA [Python]	11	11	0.80	11.86
RAPIDS cuML PCA [CUDA]	11	11	1.10	18.95
Rtsne [CPU]	9	9	1.38	4.34
FIt-SNE [CPU]	11	11	1.75	34.53
fastEmbedR t-SNE [CPU]	11	11	1.53	22.47
fastEmbedR t-SNE [CUDA]	10	10	1.51	21.20
scikit-learn t-SNE [Python]	7	7	0.76	1.12
openTSNE [Python]	10	10	1.49	21.46
RAPIDS cuML t-SNE [CUDA]	10	10	1.17	18.78
umap [CPU]	7	7	5.32	9.00
uwot [CPU]	11	11	1.68	36.66
uwot fast SGD [CPU]	11	11	1.70	36.66
fastEmbedR fuzzy UMAP [CPU]	11	11	1.50	25.21
fastEmbedR binary UMAP [CPU]	11	11	1.49	25.21
fastEmbedR fuzzy UMAP [CUDA]	10	10	1.42	21.13
fastEmbedR binary UMAP [CUDA]	10	10	1.42	21.13
umap-learn [Python]	10	10	2.09	51.67
RAPIDS cuML UMAP [CUDA]	10	10	1.47	16.00

B. CUDA device-memory increase

Method	Completed	Measured	Median	Maximum
fastEmbedR PCA [CUDA]	11	0	–	–
RAPIDS cuML PCA [CUDA]	11	0	–	–
fastEmbedR t-SNE [CUDA]	10	7	1.03	17.79
RAPIDS cuML t-SNE [CUDA]	10	7	1.28	16.97
fastEmbedR fuzzy UMAP [CUDA]	10	7	0.78	13.57
fastEmbedR binary UMAP [CUDA]	10	7	0.76	13.57
RAPIDS cuML UMAP [CUDA]	10	7	0.73	11.96

Table S10: PCA agreement with dense centered SVD on fixed subsets in campaign 20260930T203319Z. The effective rank may be below the requested rank.

Backend	Requested rank	Data sets	Score correlation	Loading cosine
CPU	2	11	0.999972	0.999998
CPU	50	11	0.999292	0.998447
CUDA	2	11	1.000000	1.000000
CUDA	50	11	1.000000	1.000000

Graph construction was saved separately from clustering. Only COIL20 and MetRef completed the latest same-graph clustering validation; the other nine data-set jobs failed, so Table S11 is restricted to six seed runs per implementation. Native CPU Louvain, Leiden, and Walktrap were compared with **igraph**; CUDA Louvain and Leiden used the corresponding native CPU memberships as a reference. Label adjusted Rand index compares the recovered communities with the data-set labels; paired adjusted Rand index compares two partitions. These are not end-to-end embedding speedups, and Metal clustering is not represented.

Table S11: Clustering in campaign 20260930T203319Z. The Runs column counts completed data-set–seed fits; times exclude graph construction. Label ARI uses external data-set labels; paired ARI compares native CPU with igraph or an accelerator with native CPU. Graph sizes identify any fixed Walktrap subset. Missing runs are not zero.

Method	Backend	Implementation	Runs	n range	Median seconds	Label ARI	Paired ARI
louvain	CPU	fastEmbedR	6	873–1440	0.006	0.496	0.838
leiden	CPU	fastEmbedR	6	873–1440	0.011	0.518	0.965
louvain	CPU	igraph	6	873–1440	0.017	0.512	0.838
leiden	CPU	igraph	6	873–1440	0.025	0.515	0.965
louvain	CUDA	fastEmbedR	6	873–1440	0.026	0.504	0.911
leiden	CUDA	fastEmbedR	6	873–1440	0.045	0.513	0.933
louvain	METAL	fastEmbedR	0	–	–	–	–
leiden	METAL	fastEmbedR	0	–	–	–	–
walktrap	CPU	fastEmbedR	6	873–1440	0.320	0.424	0.922
walktrap	CPU	igraph	6	873–1440	0.091	0.428	0.922

7. Campaign identity and unresolved gates

The source archive and the installed native library are recorded separately because this image included a container-build patch. Table S12 fixes the exact diagnostic source and benchmark identities. The campaign's final audit reported 34 failed or timed-out tasks and four output-audit failures. The replication directory retains every unsuccessful method–data-set status.

Table S12: Immutable identity of diagnostic campaign 20260930T203319Z. The campaign failed its final audit.

Item	Identifier
Package	fastEmbedR 0.1
Source commit	`7f22c52b28400535c2a246de68e7189cd9880ea1`
Source archive SHA-256	`a3bc4a09e6f60bb719cf49b5d91af876c084666ff1b390d06ea9c155d2e991b9`
Build patch SHA-256	`8a725d2d2579425ff8d7edd3684f74c3fbb9ec1e5e3624c53e3eb3280b1188c7`
Container SHA-256	`54e9604d951ecb3d80f49671d9db7a35cf19f1342eb3ee7c708cbc0c3a7f0aa8`
Installed library SHA-256	`d44f08b574a4932bd36f7afb1d0da0895ccaaa9e1e5ca3f4c040ef2357d649fc`
Benchmark manifest SHA-256	`d0dddde1d82fa3faa01a9617db55b4128f8a04de61b8fd4dfb6f117d99e6f44c`

The experimental file-backed phase was added after this image and campaign were created. No BigANN 1-billion-row or two-GPU result from that phase appears in these tables. Its six planned analyses are matched correctness, real-data scaling, memory boundary, full versus landmark, multi-GPU, and checkpoint recovery. The phase must be rerun with source and output checksums before any result is included in a release-level comparison.

8. Embedding gallery

The gallery uses coordinates from campaign 20260929T210526Z, separate from the full-data timing campaign in Table S1. Only seed-4 layouts are shown. Other seeds contribute to numerical summaries but are not illustrated. That campaign also failed its audit; these plots are visual diagnostics, not release-level validation. Missing panels are labelled and are never filled from another campaign.

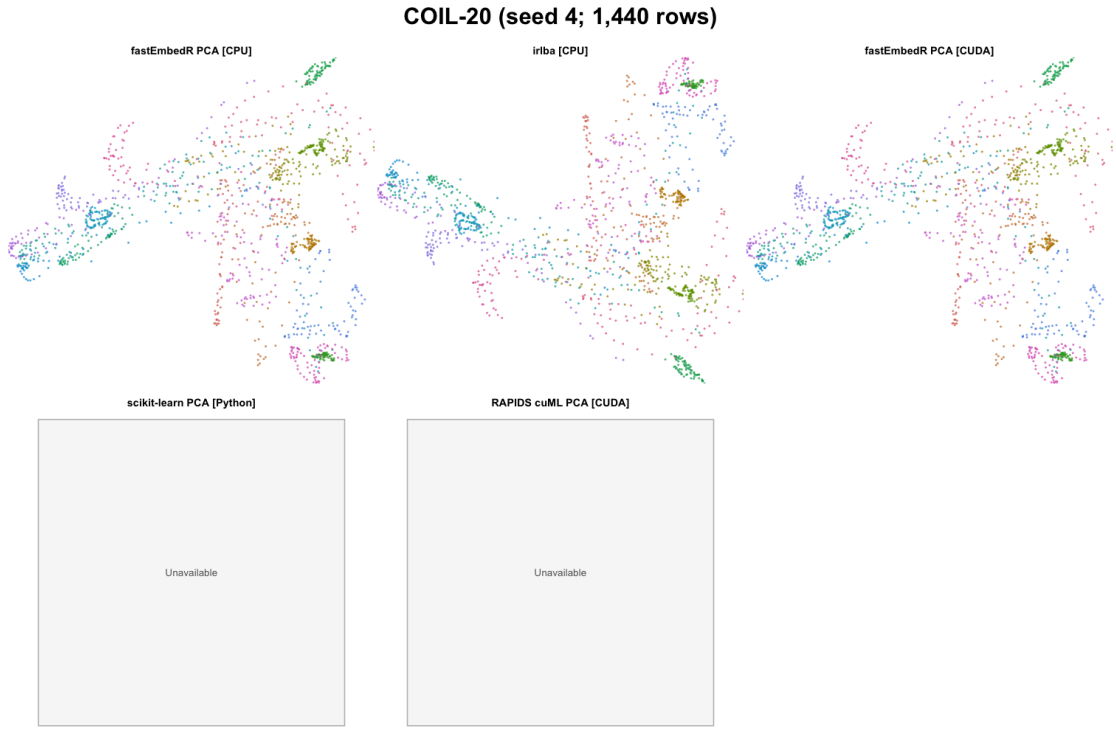

Figure S1: COIL-20 PCA outputs for the tested methods (seed 4; 1,440 rows). Colors encode benchmark labels and were not used for fitting.

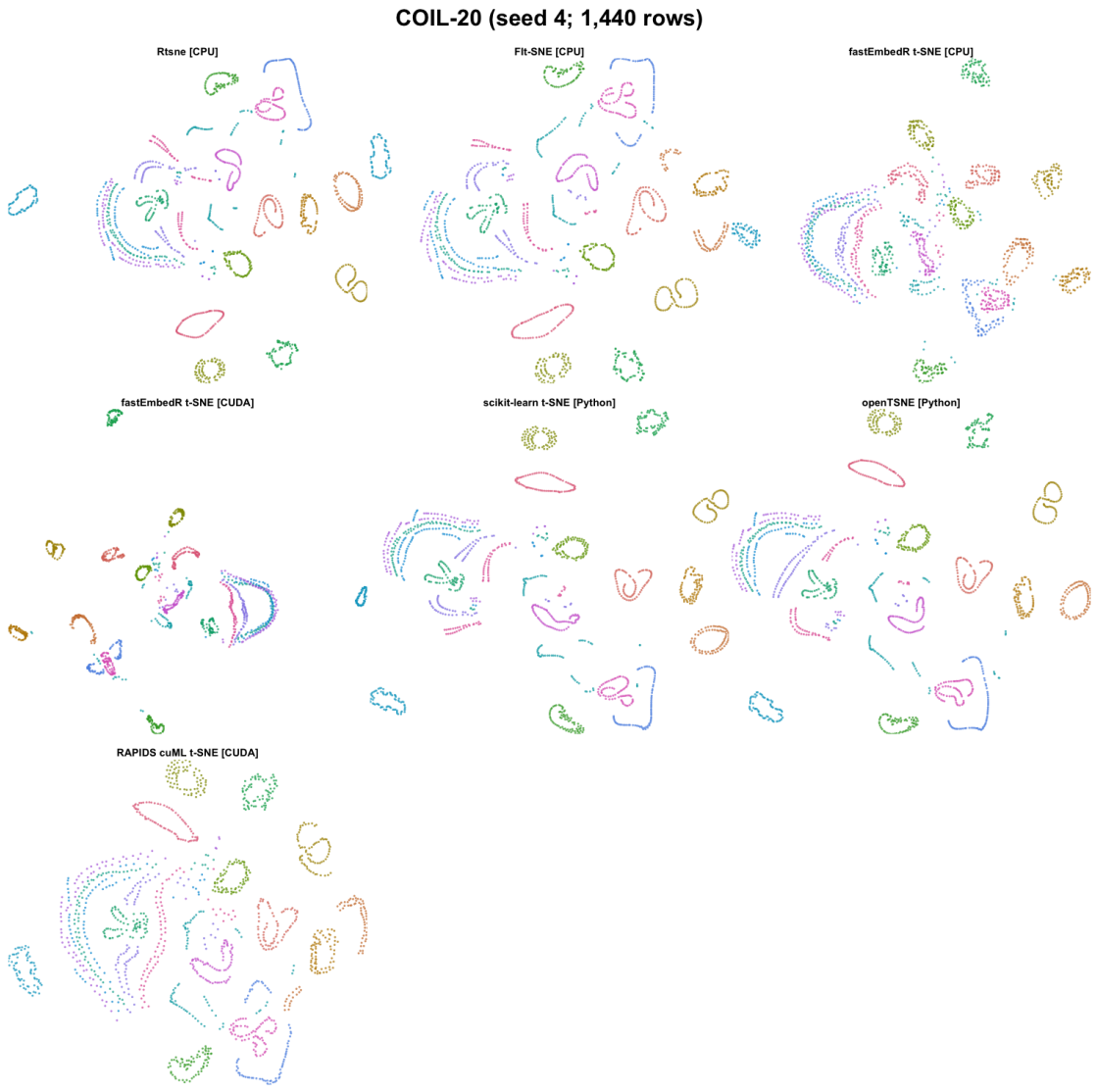

Figure S2: COIL-20 t-SNE outputs for the tested methods (seed 4; 1,440 rows). Colors encode benchmark labels and were not used for fitting.

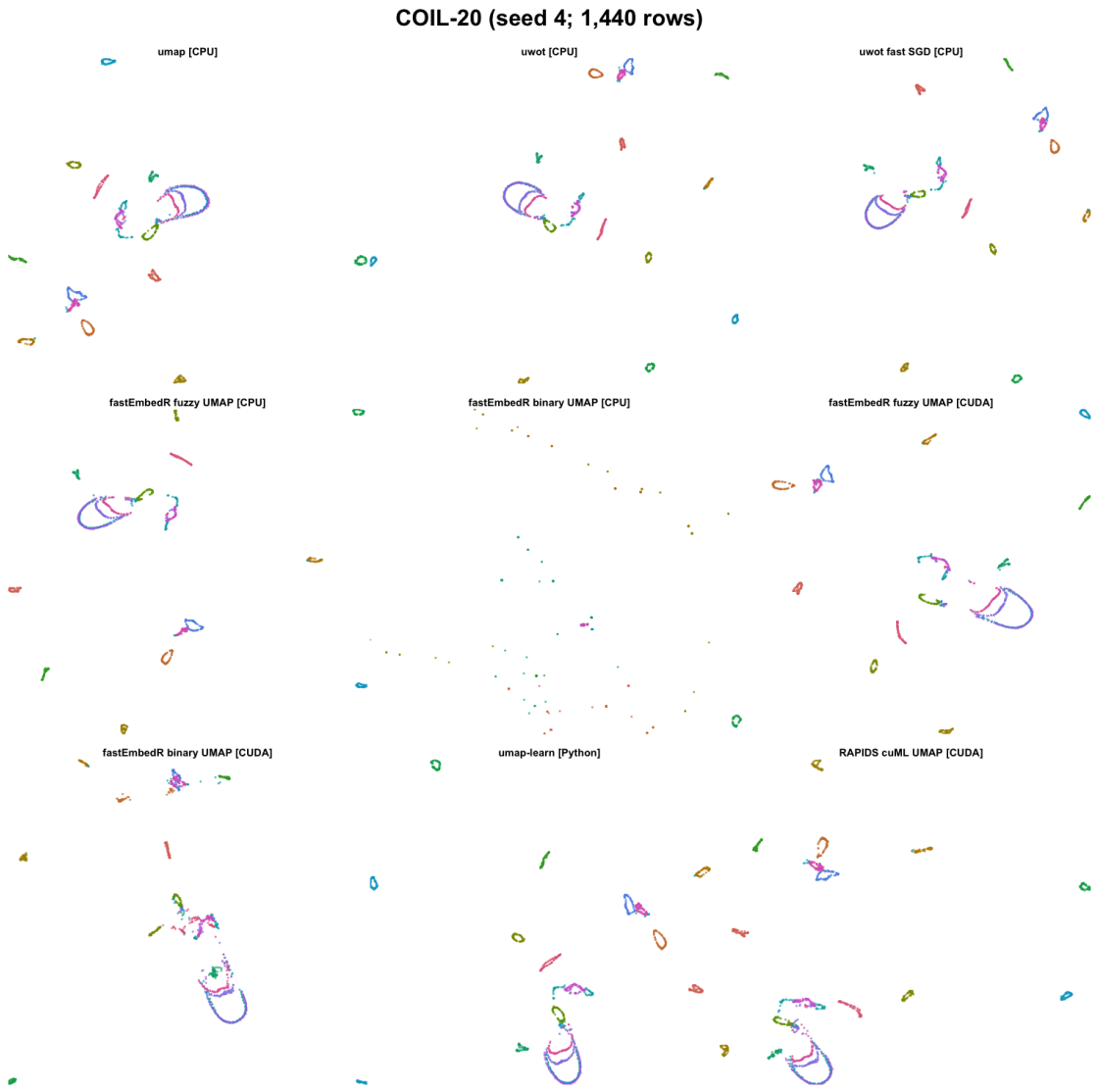

Figure S3: COIL-20 UMAP outputs for the tested methods (seed 4; 1,440 rows). Colors encode benchmark labels and were not used for fitting.

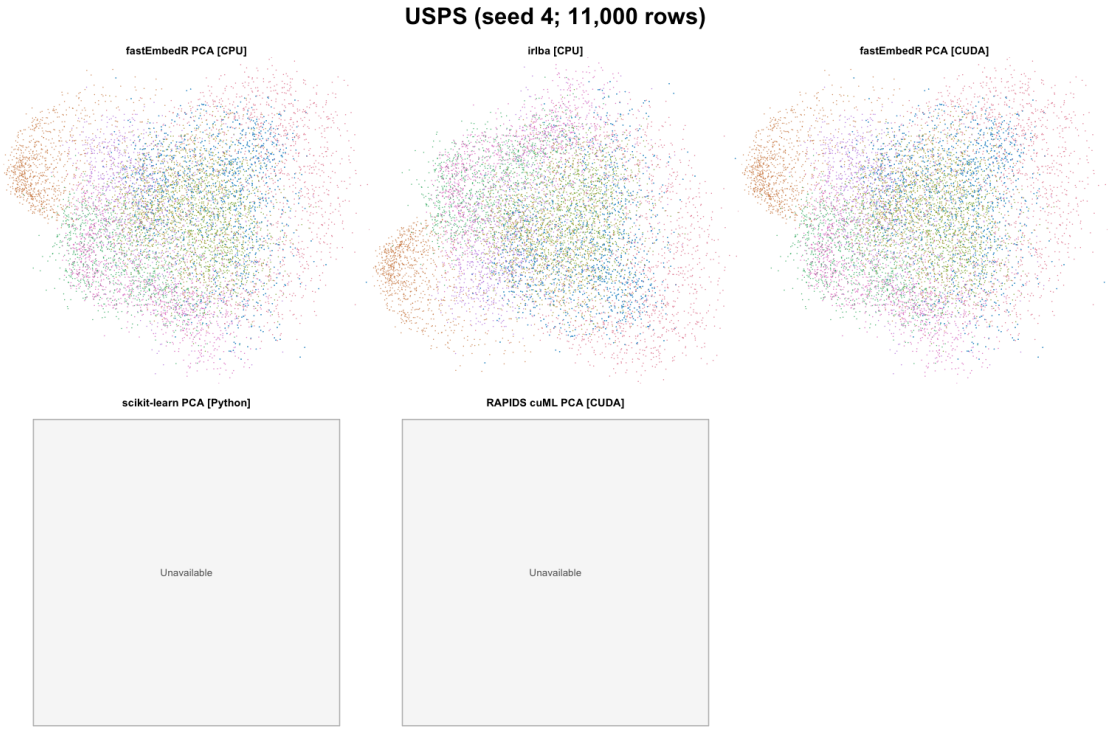

Figure S4: USPS PCA outputs for the tested methods (seed 4; 11,000 rows). Colors encode benchmark labels and were not used for fitting.

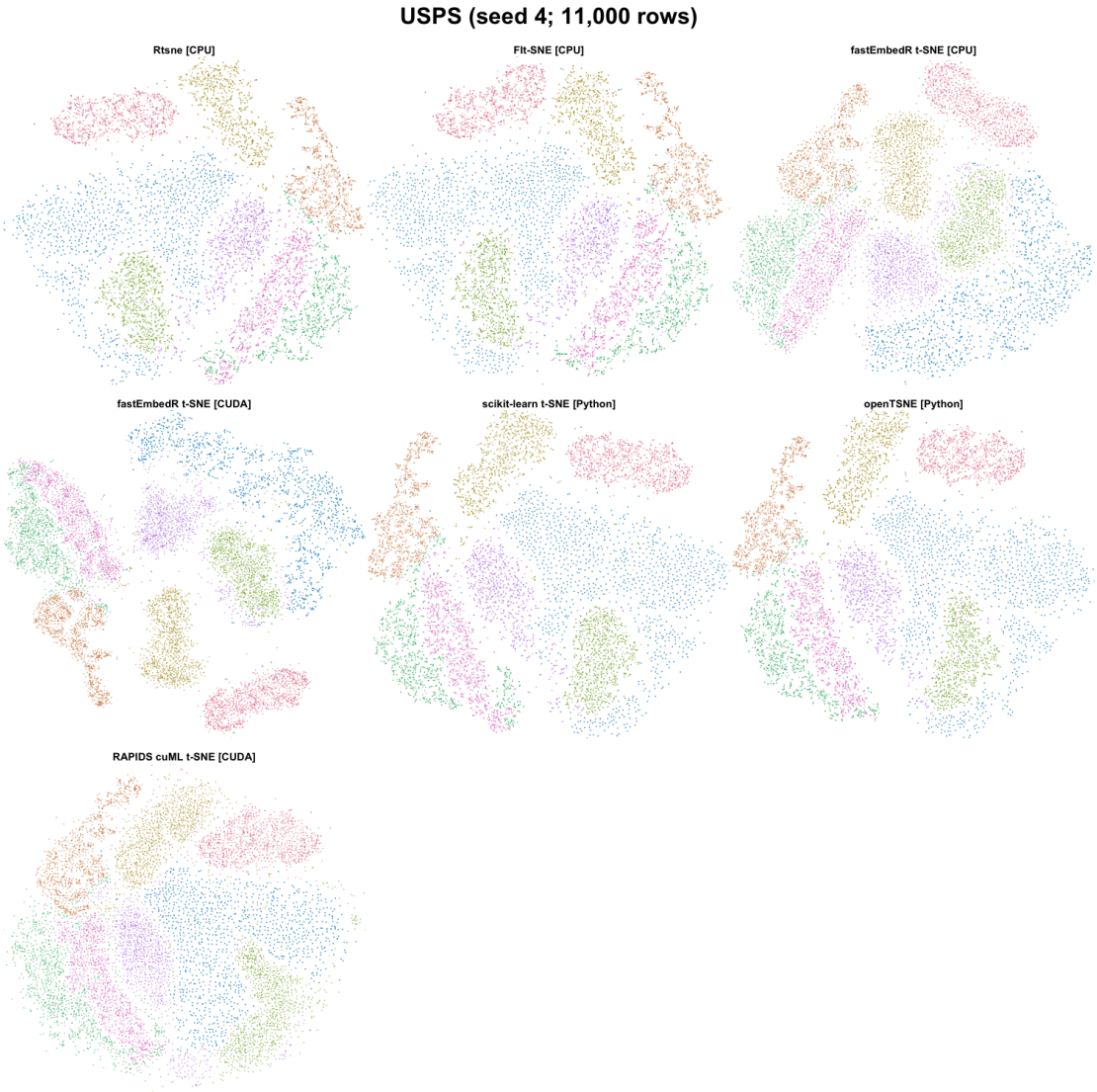

Figure S5: USPS t-SNE outputs for the tested methods (seed 4; 11,000 rows). Colors encode benchmark labels and were not used for fitting.

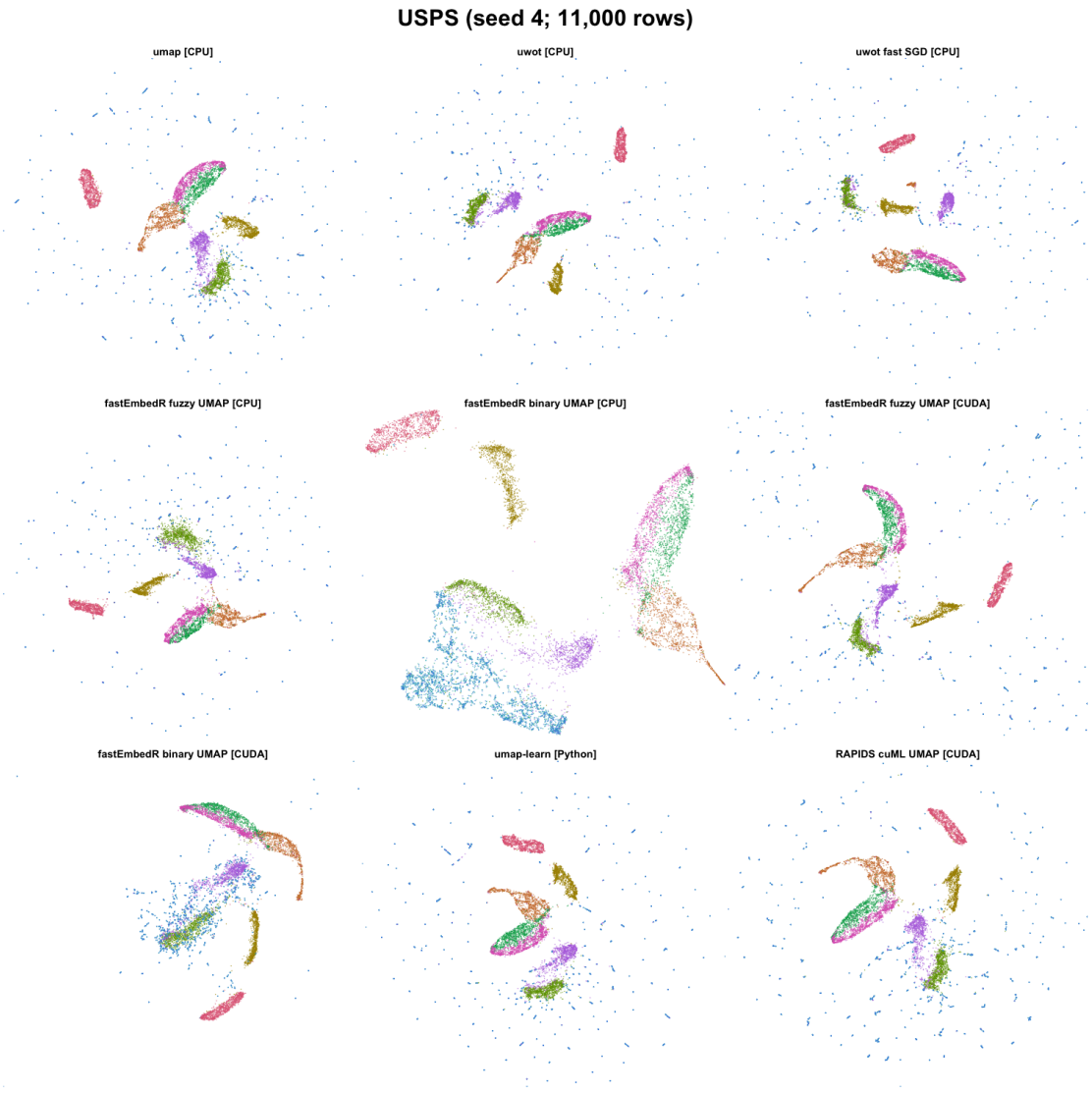

Figure S6: USPS UMAP outputs for the tested methods (seed 4; 11,000 rows). Colors encode benchmark labels and were not used for fitting.

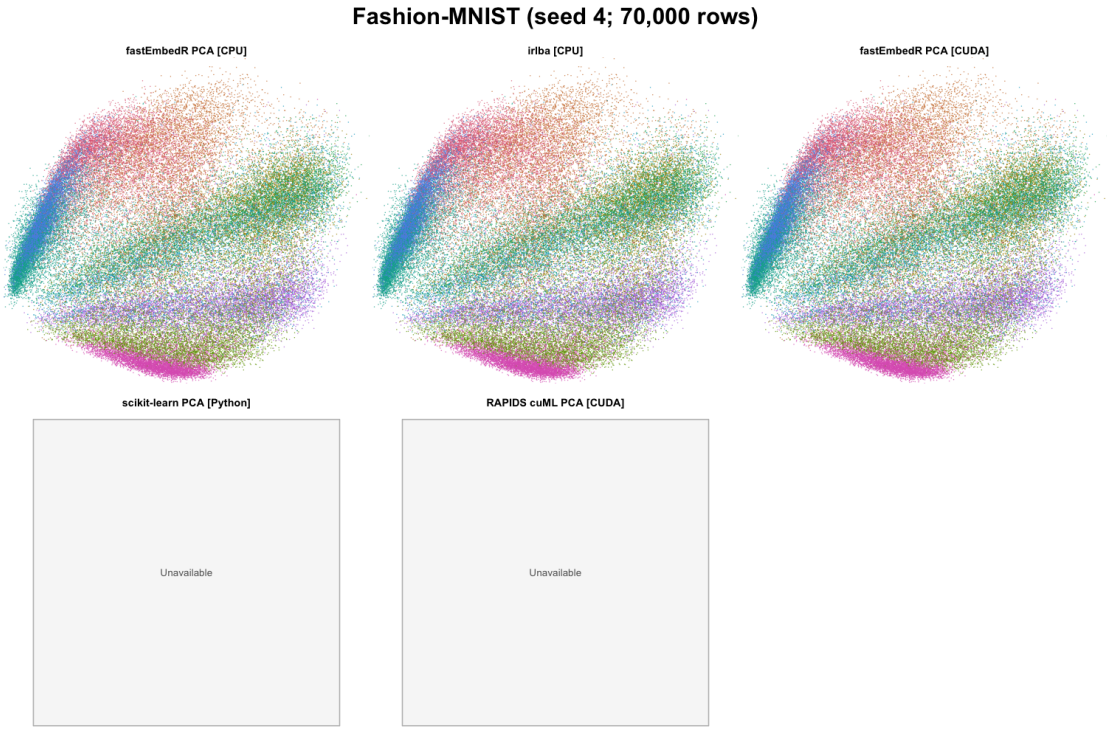

Figure S7: Fashion-MNIST PCA outputs for the tested methods (seed 4; 70,000 rows). Colors encode benchmark labels and were not used for fitting.

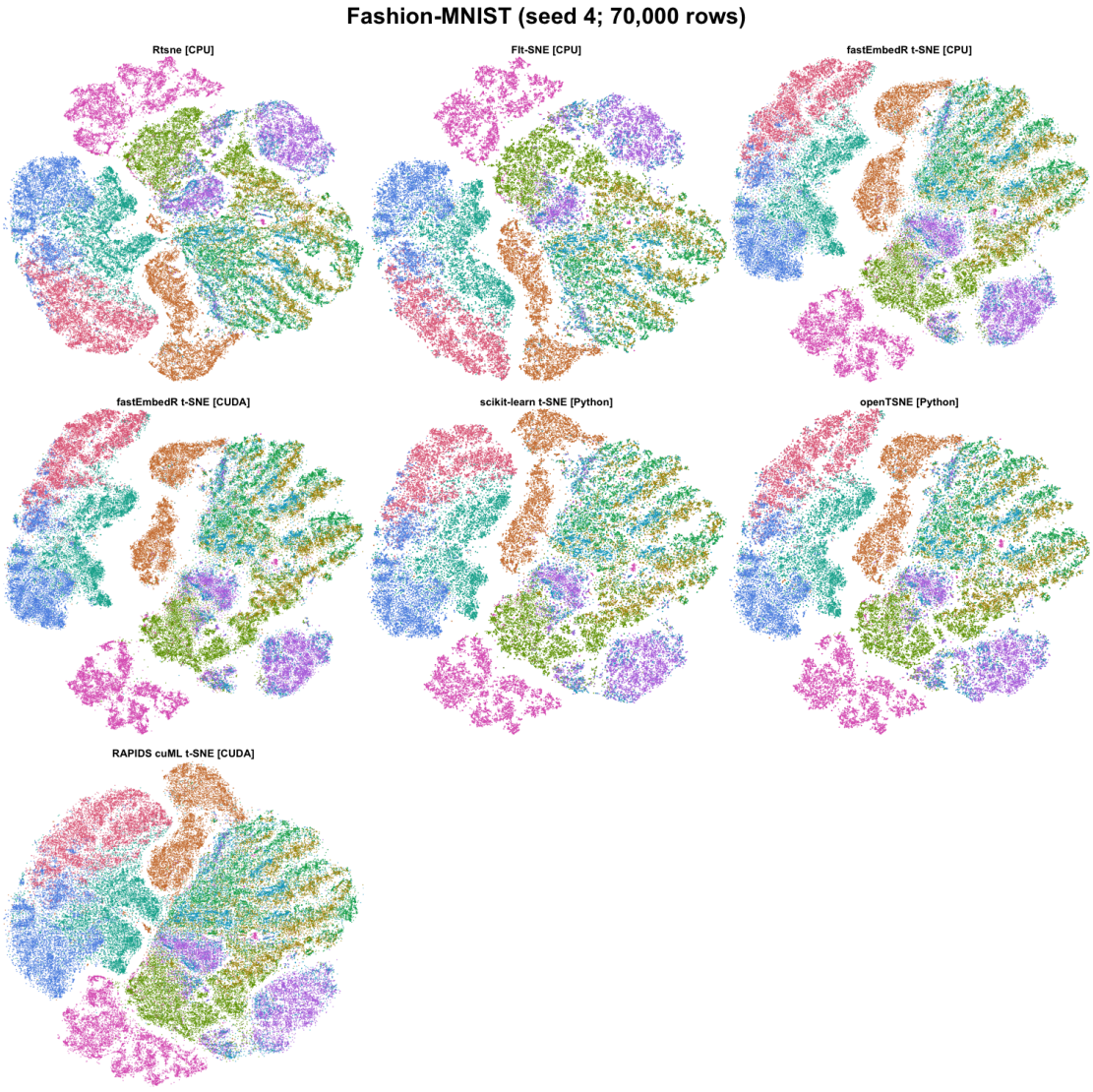

Figure S8: Fashion-MNIST t-SNE outputs for the tested methods (seed 4; 70,000 rows). Colors encode benchmark labels and were not used for fitting.

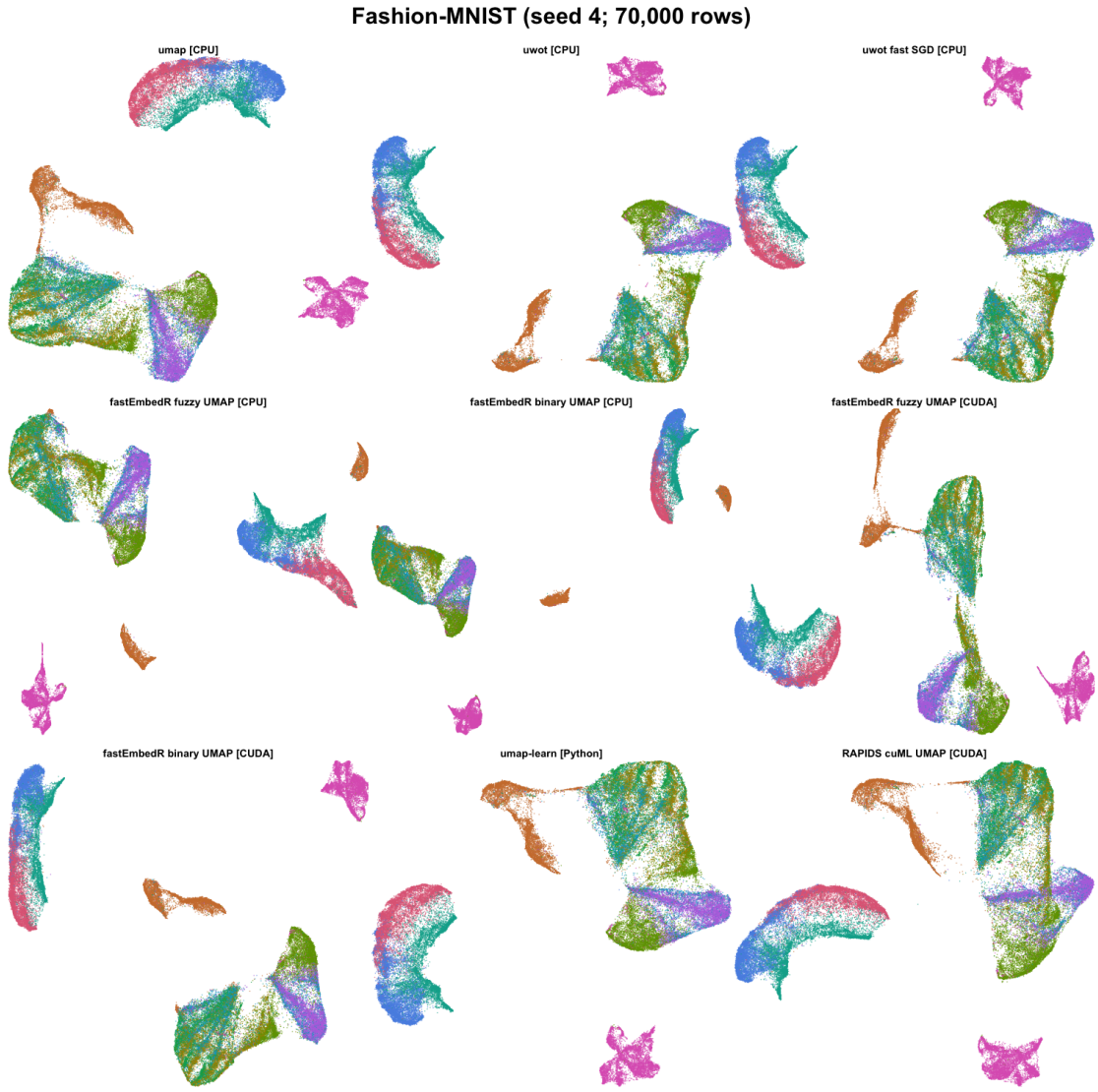

Figure S9: Fashion-MNIST UMAP outputs for the tested methods (seed 4; 70,000 rows). Colors encode benchmark labels and were not used for fitting.

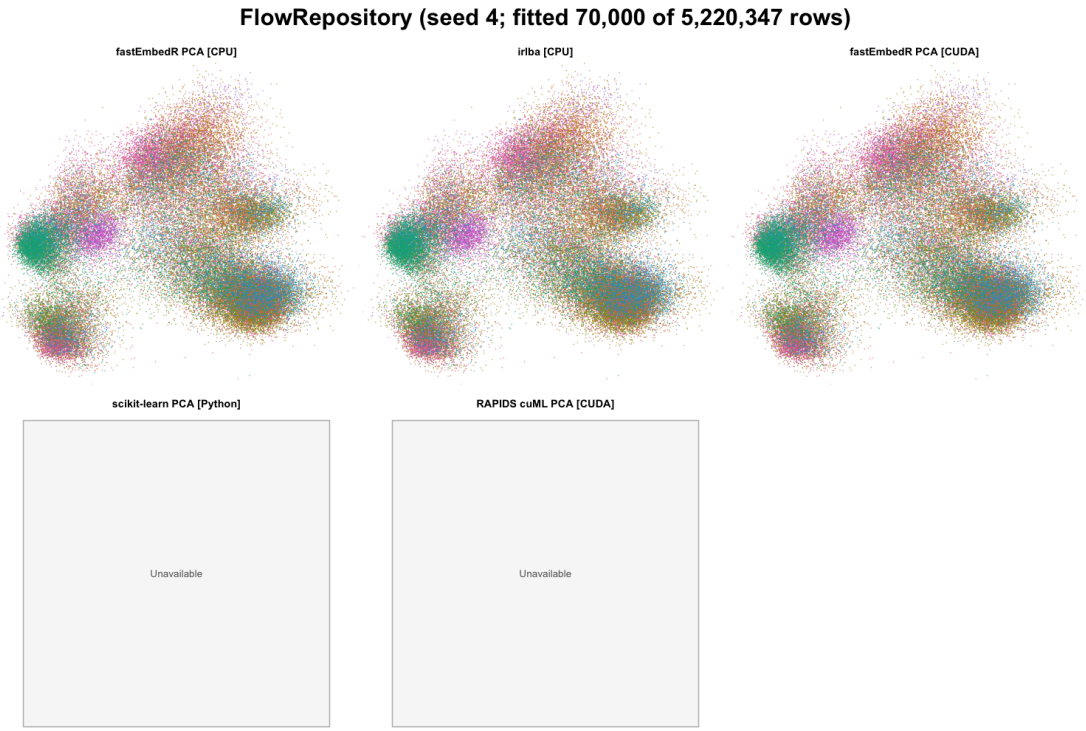

Figure S10: FlowRepository PCA outputs for the tested methods (seed 4; fitted 70,000 of 5,220,347 rows). Colors encode benchmark labels and were not used for fitting.

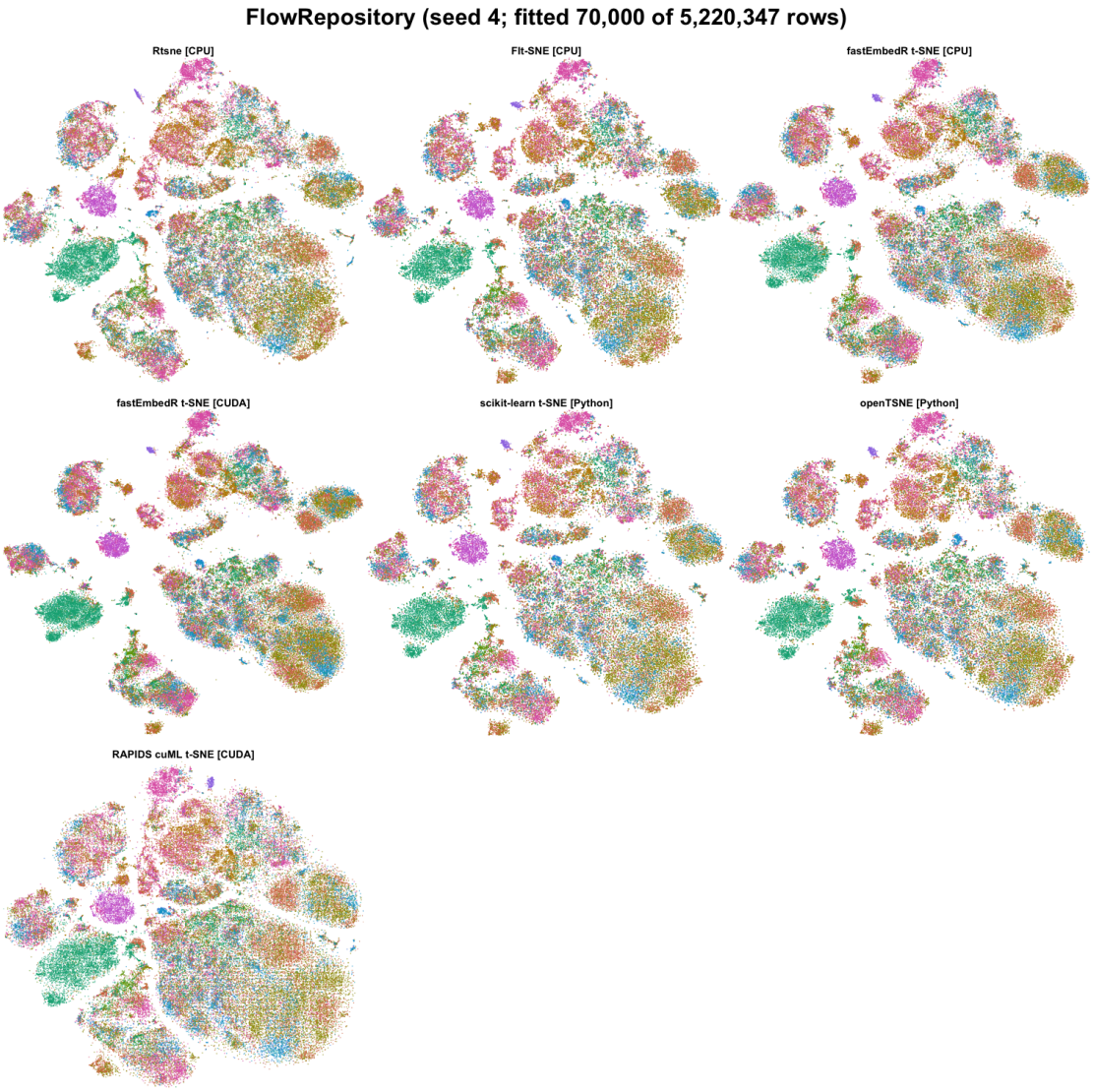

Figure S11: FlowRepository t-SNE outputs for the tested methods (seed 4; fitted 70,000 of 5,220,347 rows). Colors encode benchmark labels and were not used for fitting.

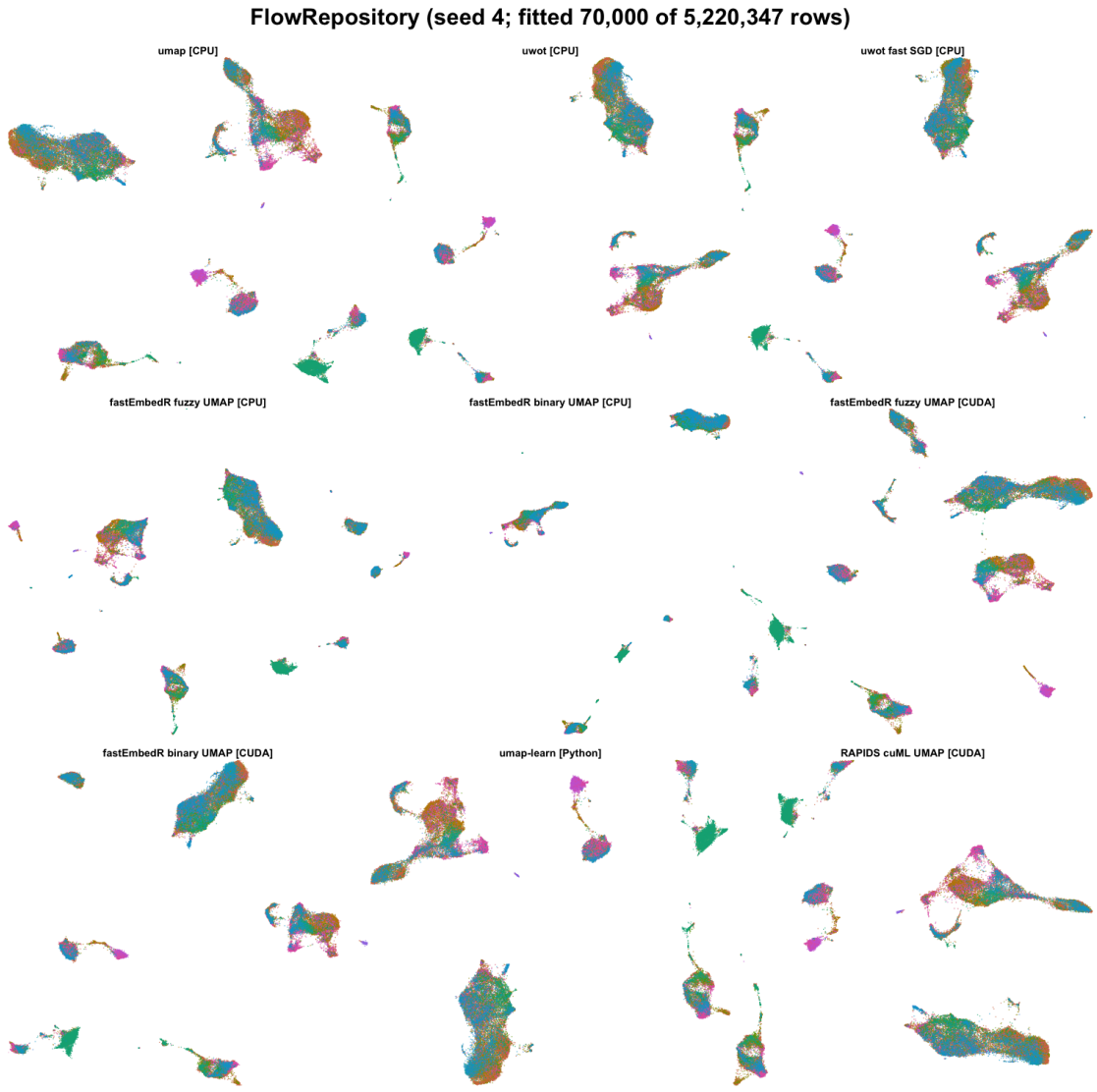

Figure S12: FlowRepository UMAP outputs for the tested methods (seed 4; fitted 70,000 of 5,220,347 rows). Colors encode benchmark labels and were not used for fitting.

Figure S13: flow18 PCA outputs for the tested methods (seed 4; fitted 70,000 of 1,000,021 rows). Colors encode benchmark labels and were not used for fitting.

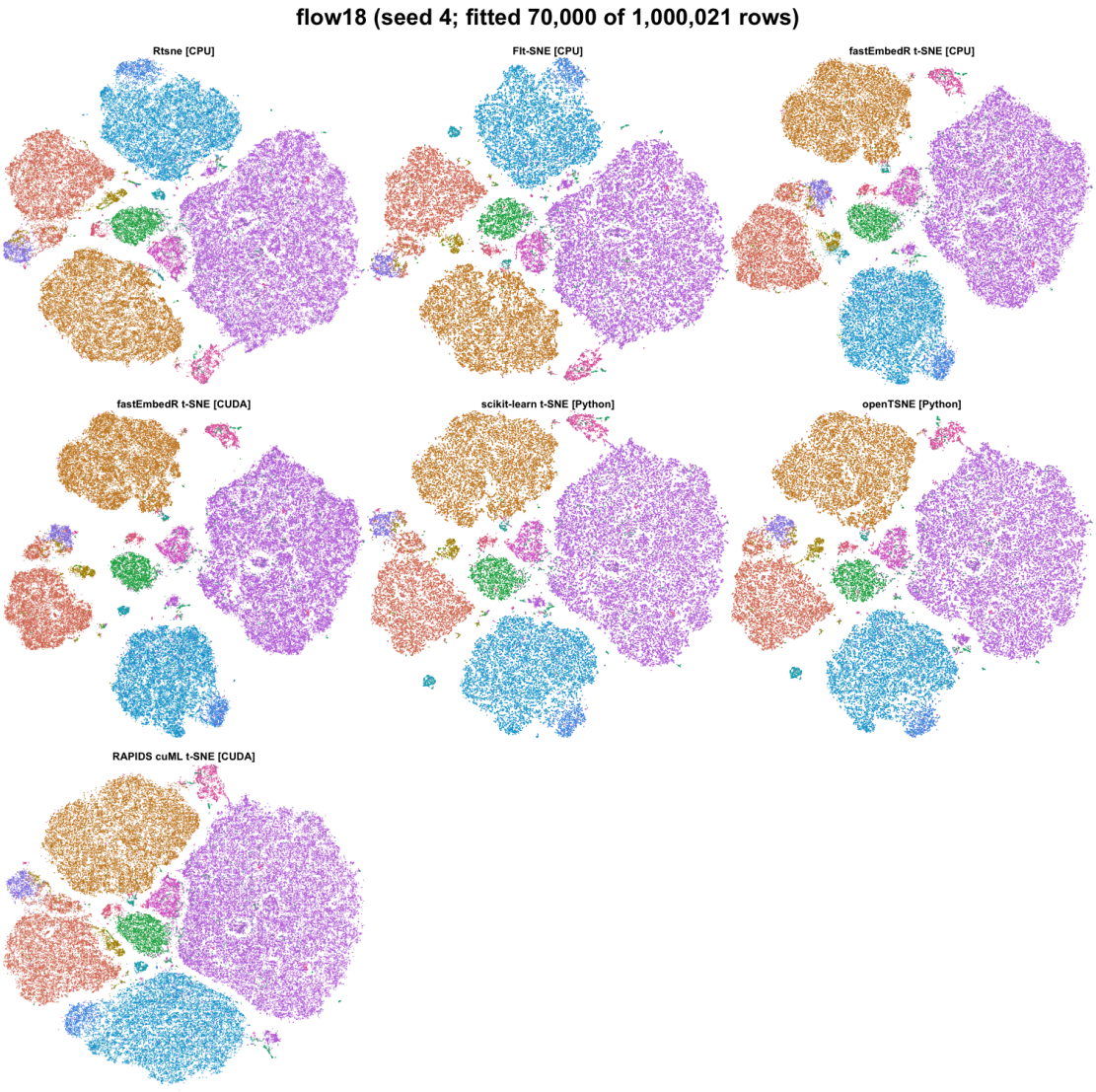

Figure S14: flow18 t-SNE outputs for the tested methods (seed 4; fitted 70,000 of 1,000,021 rows). Colors encode benchmark labels and were not used for fitting.

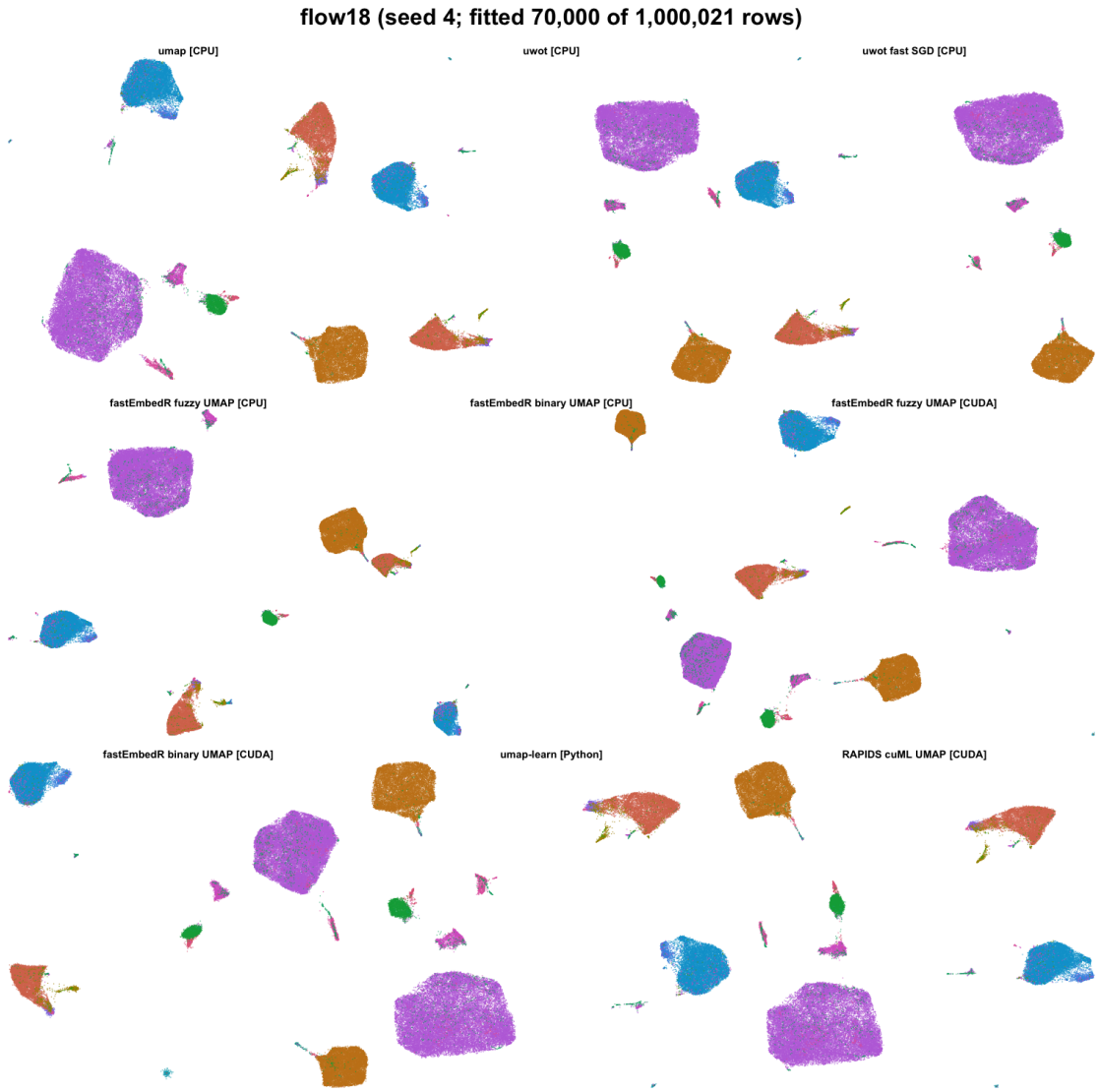

Figure S15: flow18 UMAP outputs for the tested methods (seed 4; fitted 70,000 of 1,000,021 rows). Colors encode benchmark labels and were not used for fitting.

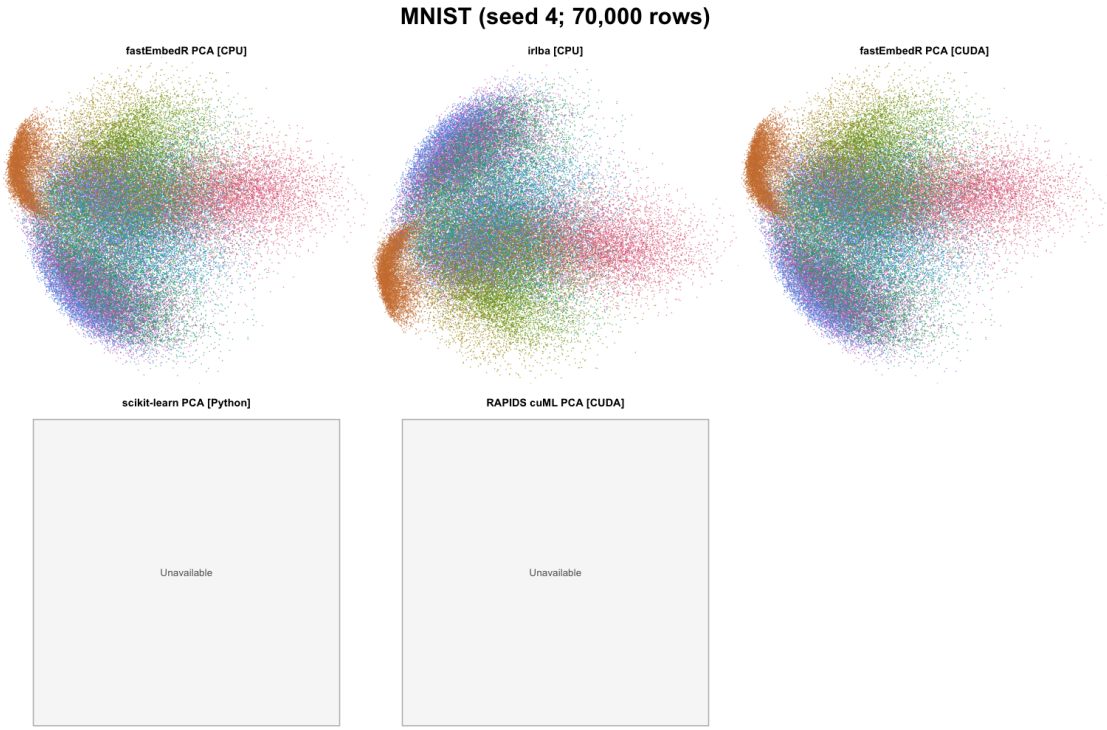

Figure S16: MNIST PCA outputs for the tested methods (seed 4; 70,000 rows). Colors encode benchmark labels and were not used for fitting.

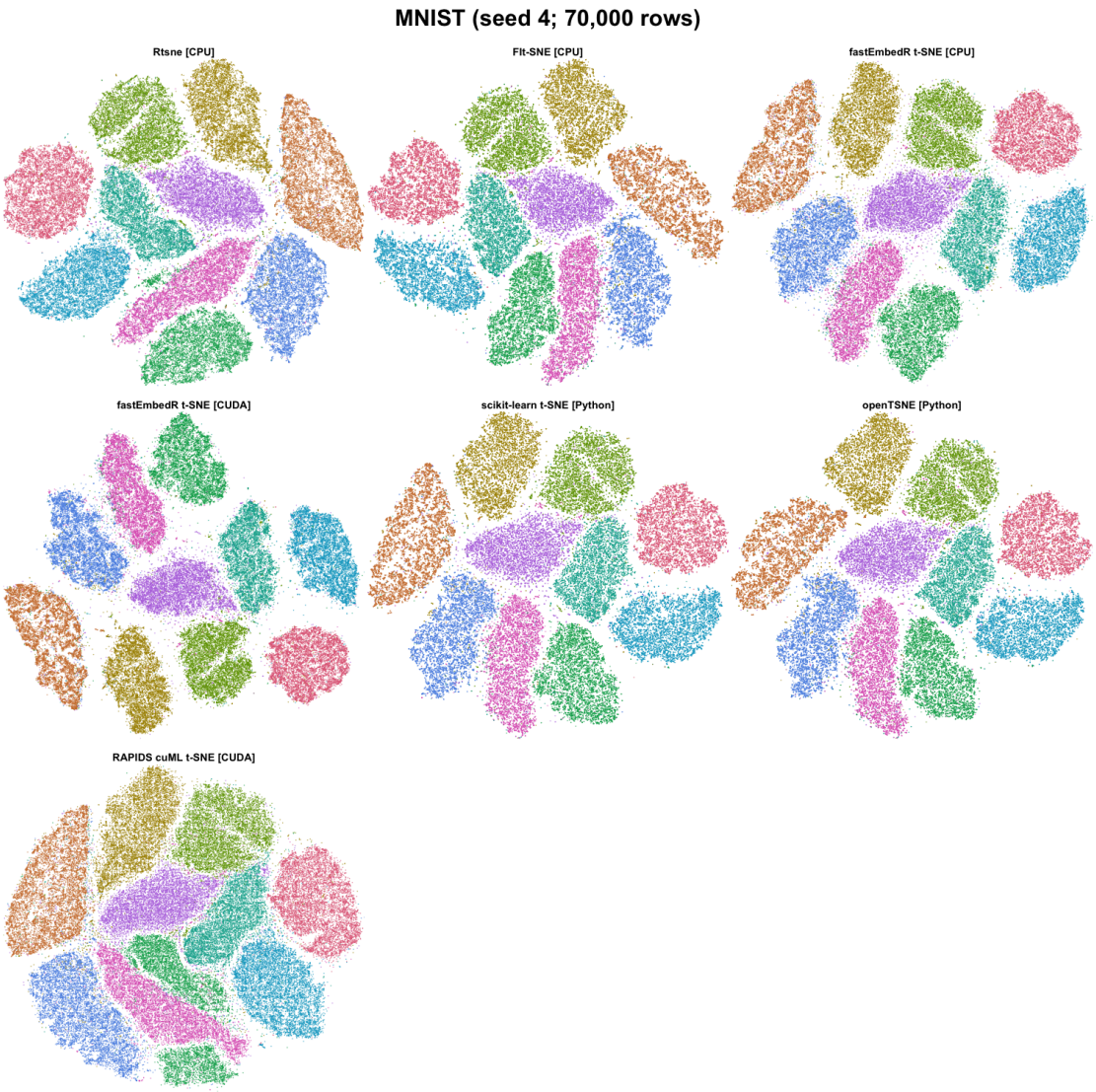

Figure S17: MNIST t-SNE outputs for the tested methods (seed 4; 70,000 rows). Colors encode benchmark labels and were not used for fitting.

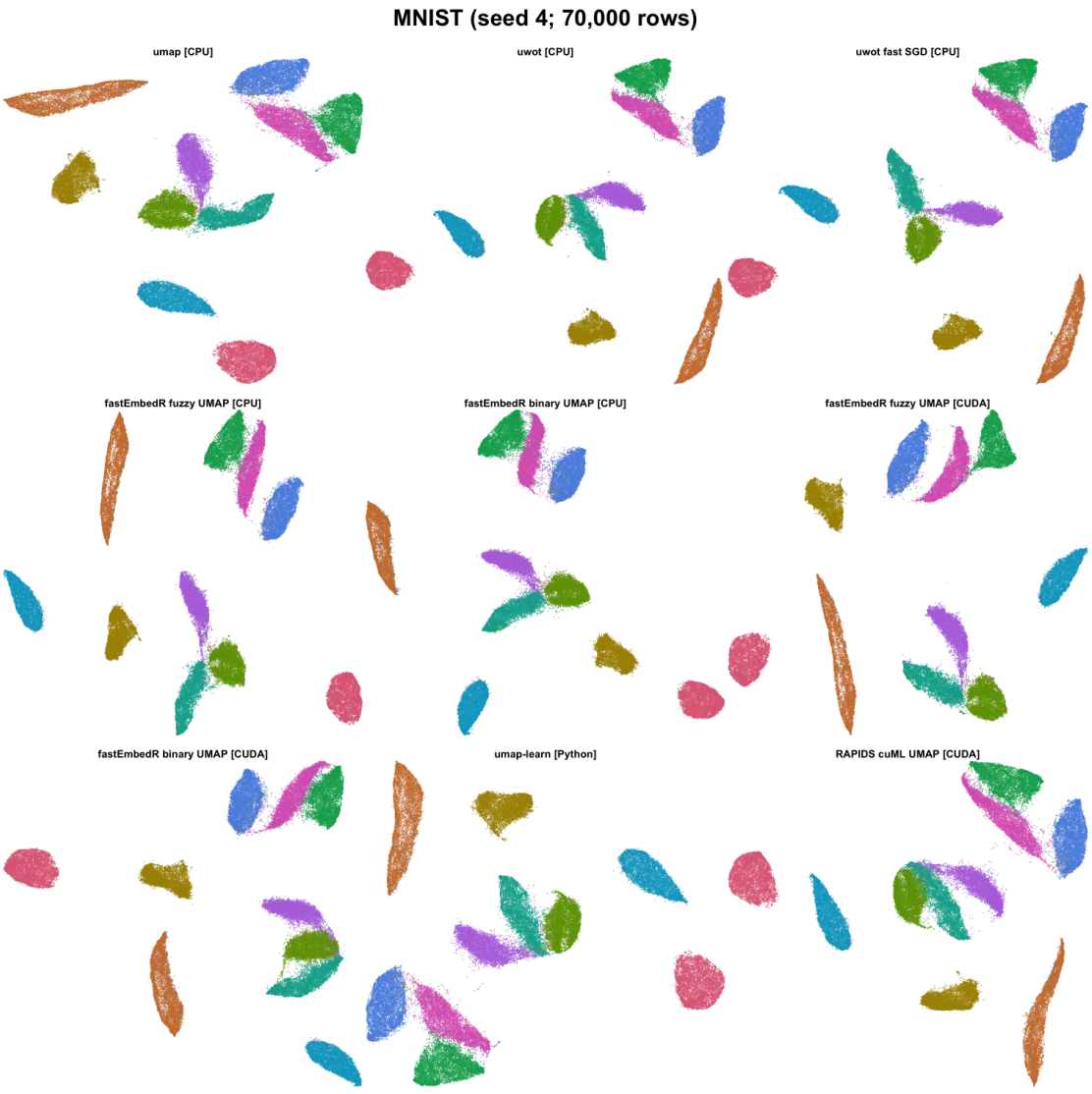

Figure S18: MNIST UMAP outputs for the tested methods (seed 4; 70,000 rows). Colors encode benchmark labels and were not used for fitting.

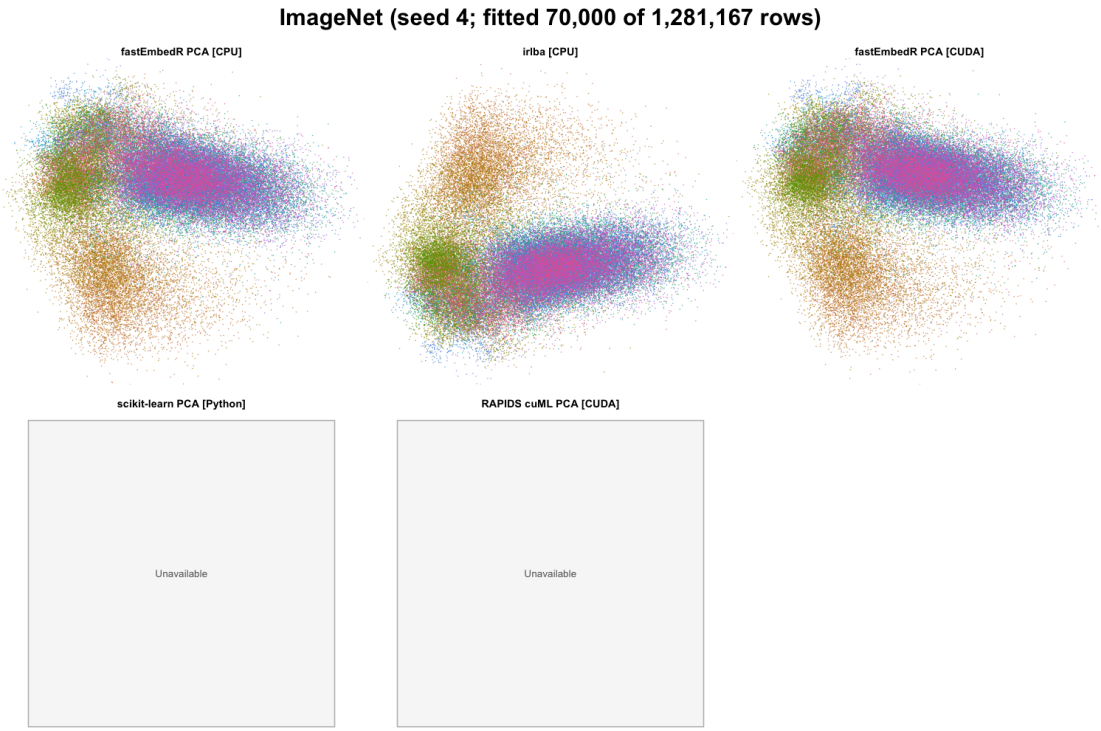

Figure S19: ImageNet PCA outputs for the tested methods (seed 4; fitted 70,000 of 1,281,167 rows). Colors encode benchmark labels and were not used for fitting.

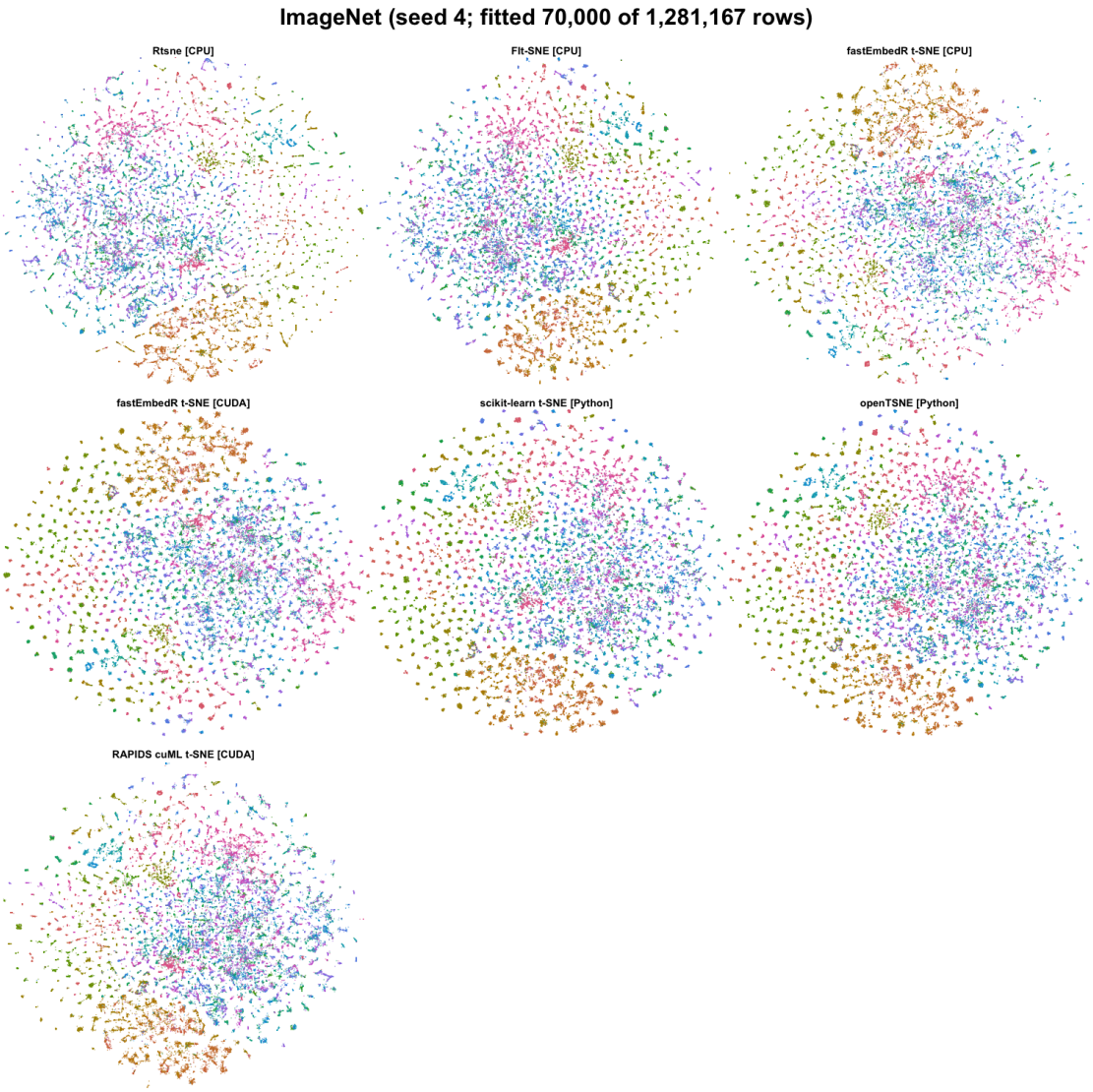

Figure S20: ImageNet t-SNE outputs for the tested methods (seed 4; fitted 70,000 of 1,281,167 rows). Colors encode benchmark labels and were not used for fitting.

Figure S21: ImageNet UMAP outputs for the tested methods (seed 4; fitted 70,000 of 1,281,167 rows). Colors encode benchmark labels and were not used for fitting.

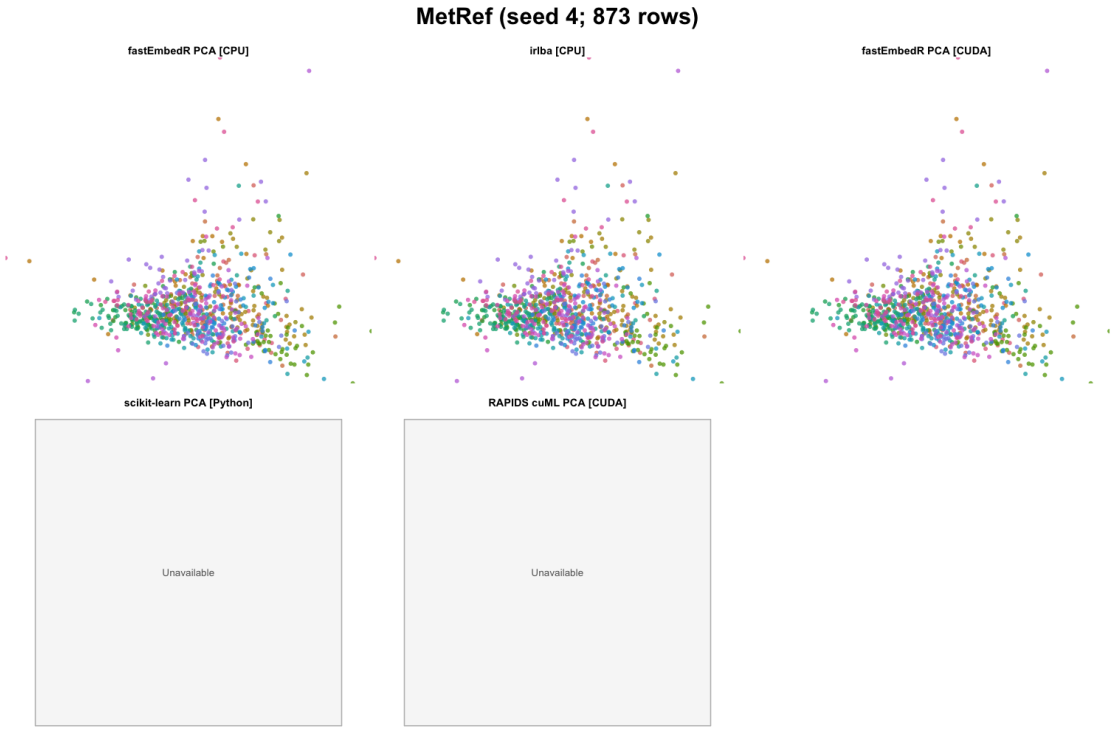

Figure S22: MetRef PCA outputs for the tested methods (seed 4; 873 rows). Colors encode benchmark labels and were not used for fitting.

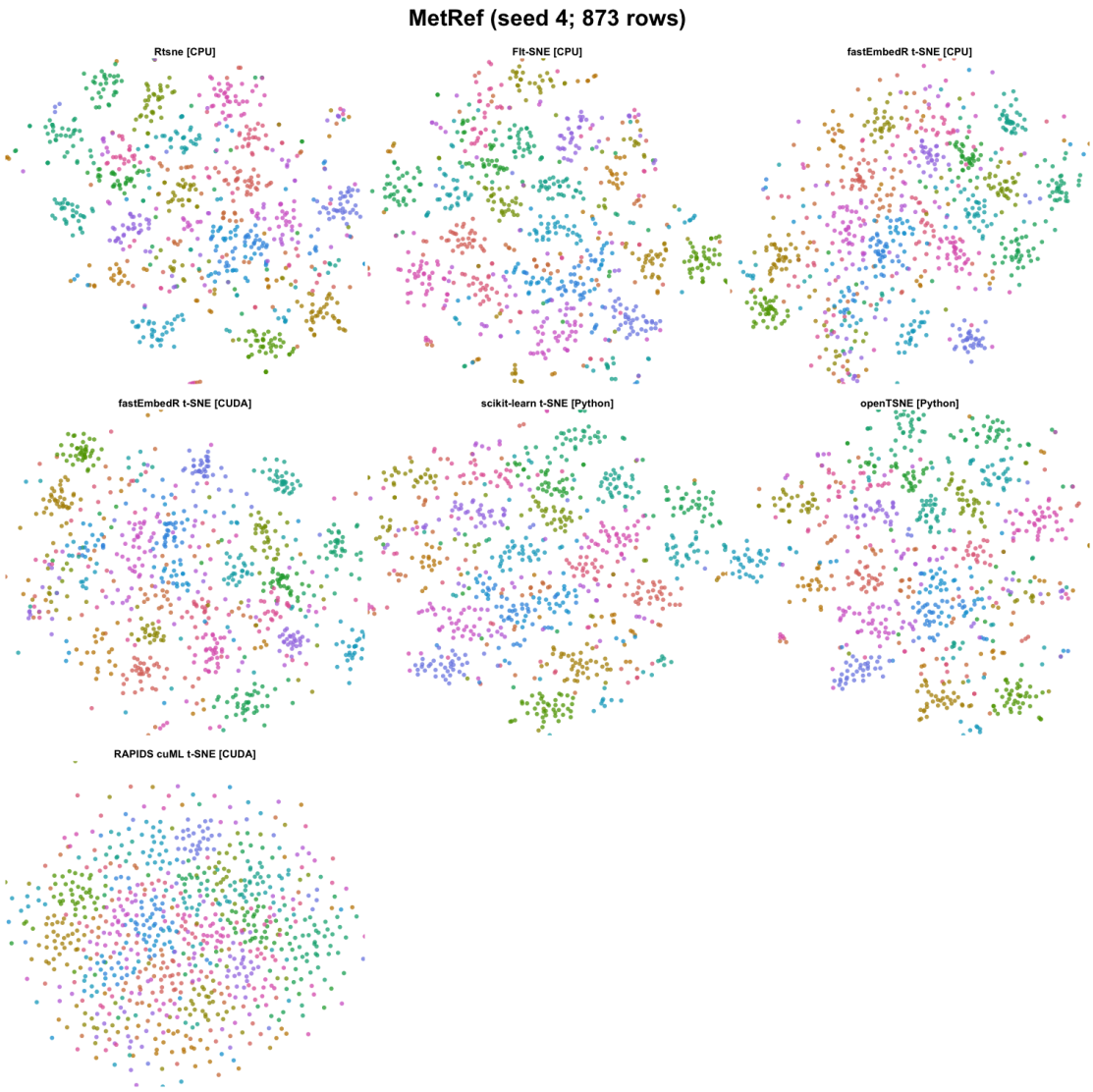

Figure S23: MetRef t-SNE outputs for the tested methods (seed 4; 873 rows). Colors encode benchmark labels and were not used for fitting.

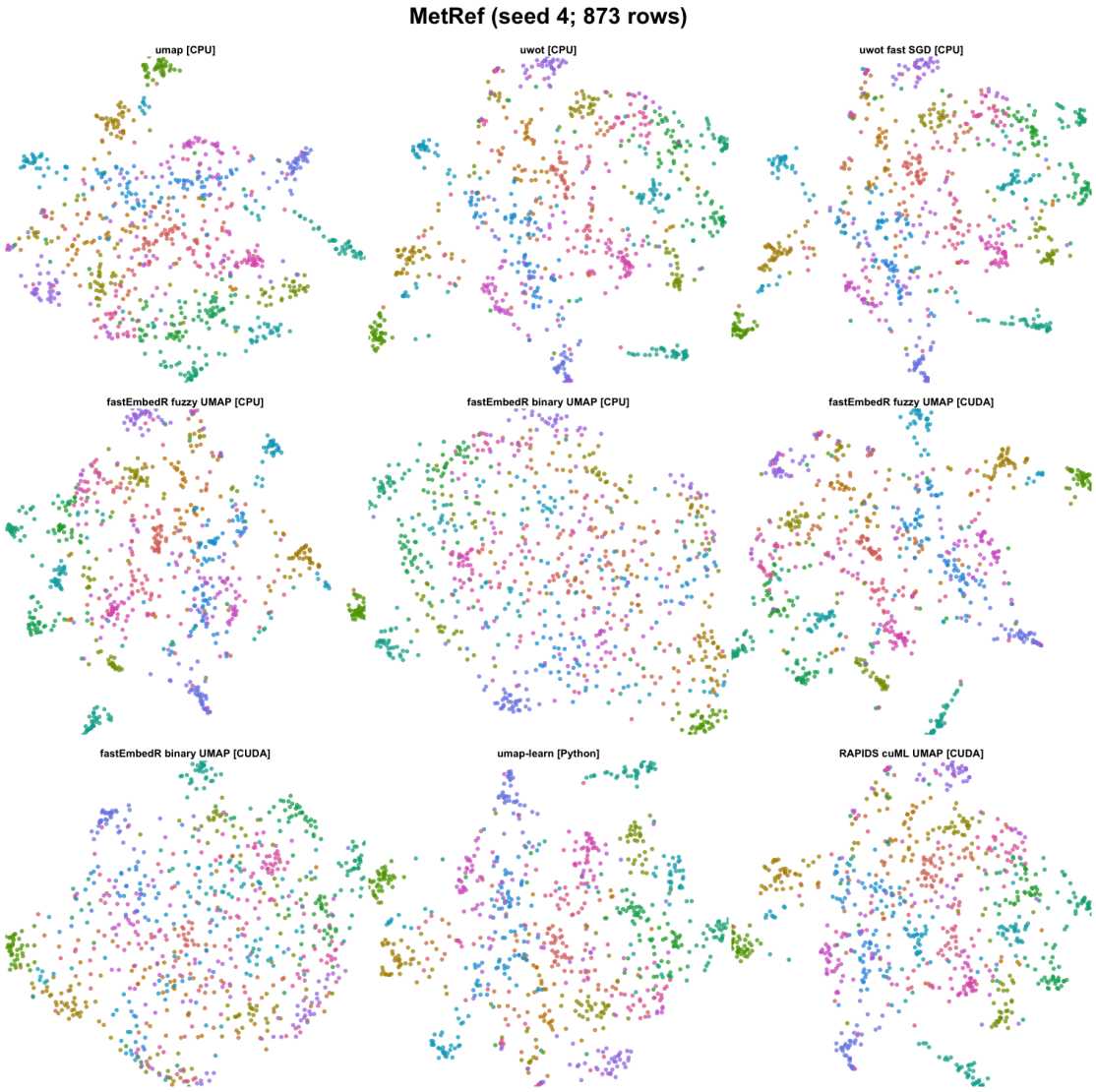

Figure S24: MetRef UMAP outputs for the tested methods (seed 4; 873 rows). Colors encode benchmark labels and were not used for fitting.

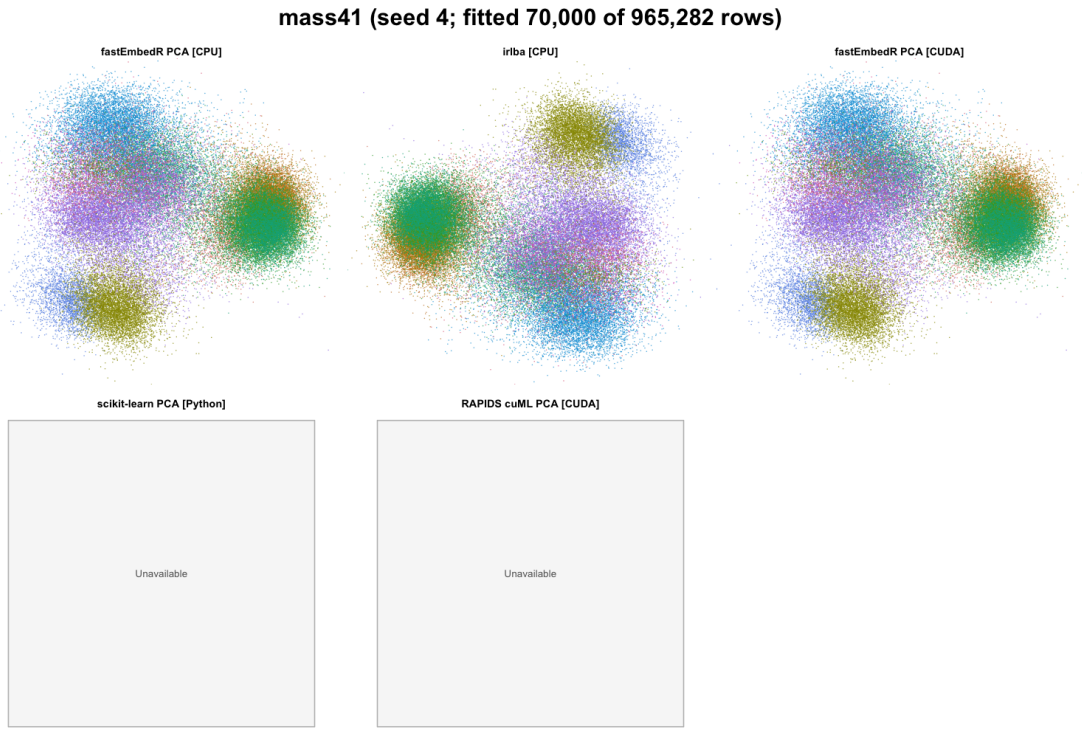

Figure S25: mass41 PCA outputs for the tested methods (seed 4; fitted 70,000 of 965,282 rows). Colors encode benchmark labels and were not used for fitting.

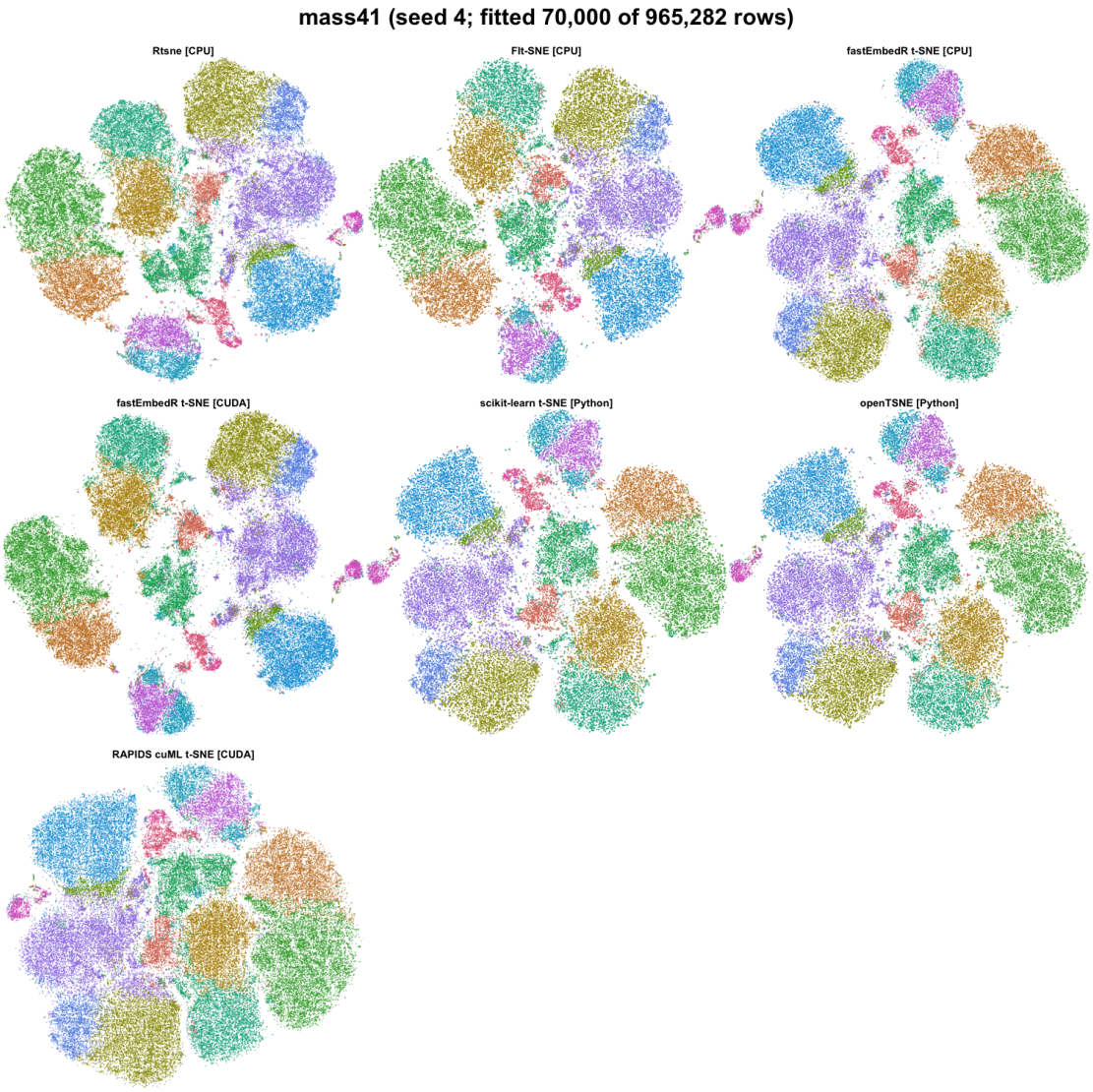

Figure S26: mass41 t-SNE outputs for the tested methods (seed 4; fitted 70,000 of 965,282 rows). Colors encode benchmark labels and were not used for fitting.

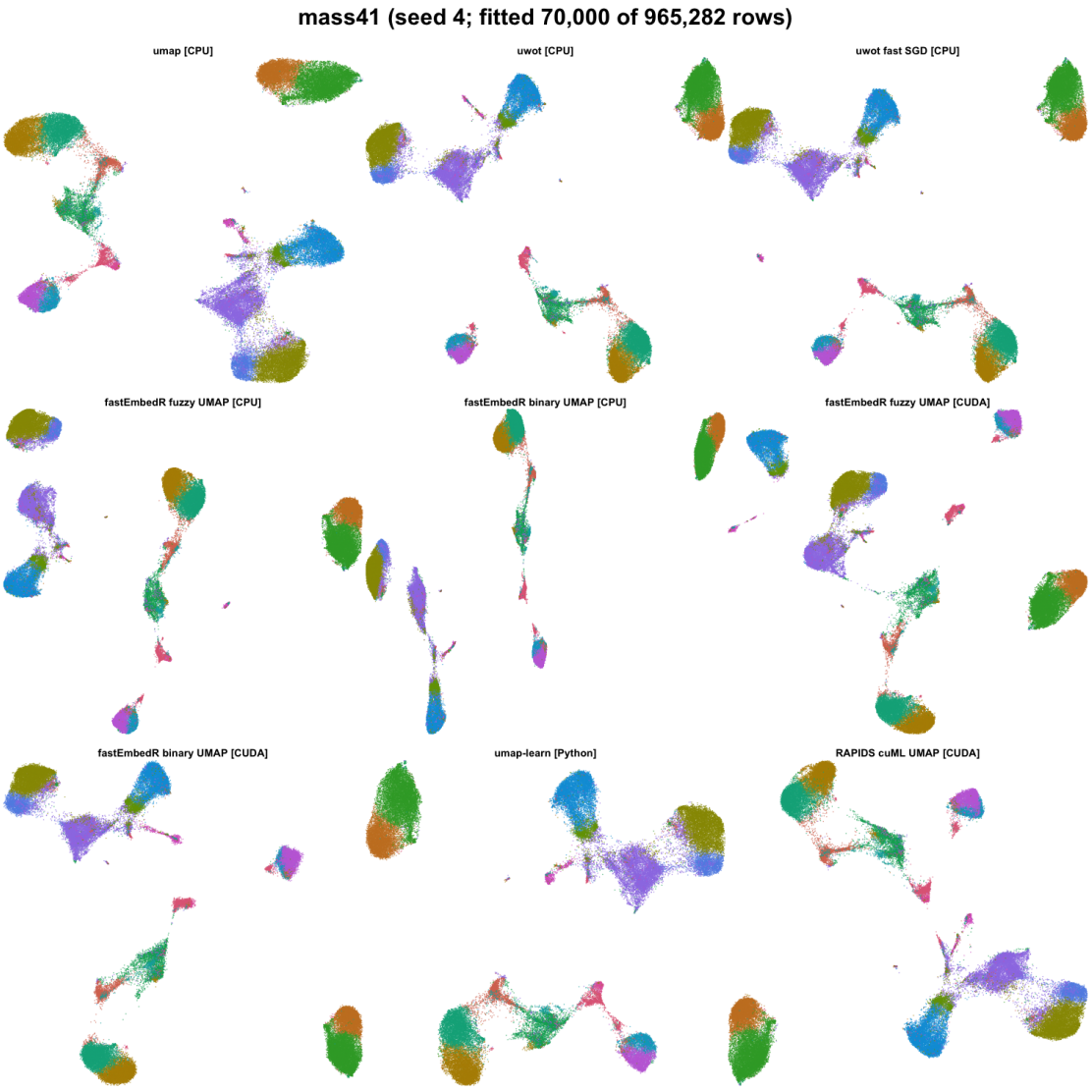

Figure S27: mass41 UMAP outputs for the tested methods (seed 4; fitted 70,000 of 965,282 rows). Colors encode benchmark labels and were not used for fitting.

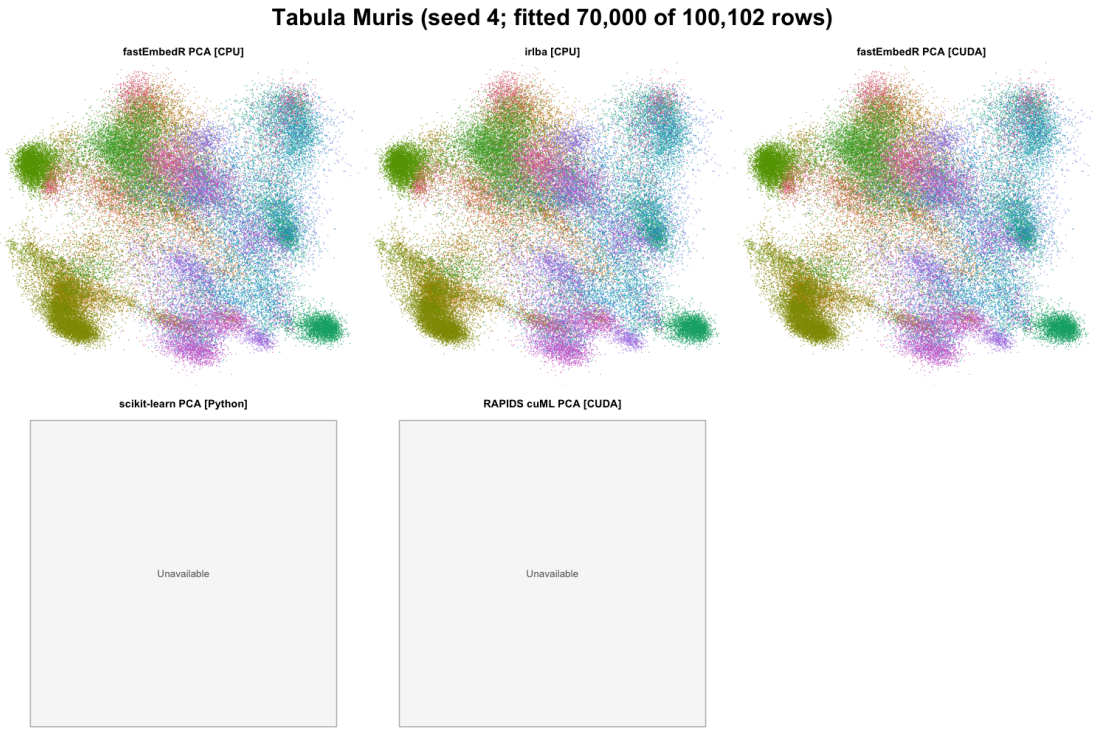

Figure S28: Tabula Muris PCA outputs for the tested methods (seed 4; fitted 70,000 of 100,102 rows). Colors encode benchmark labels and were not used for fitting.

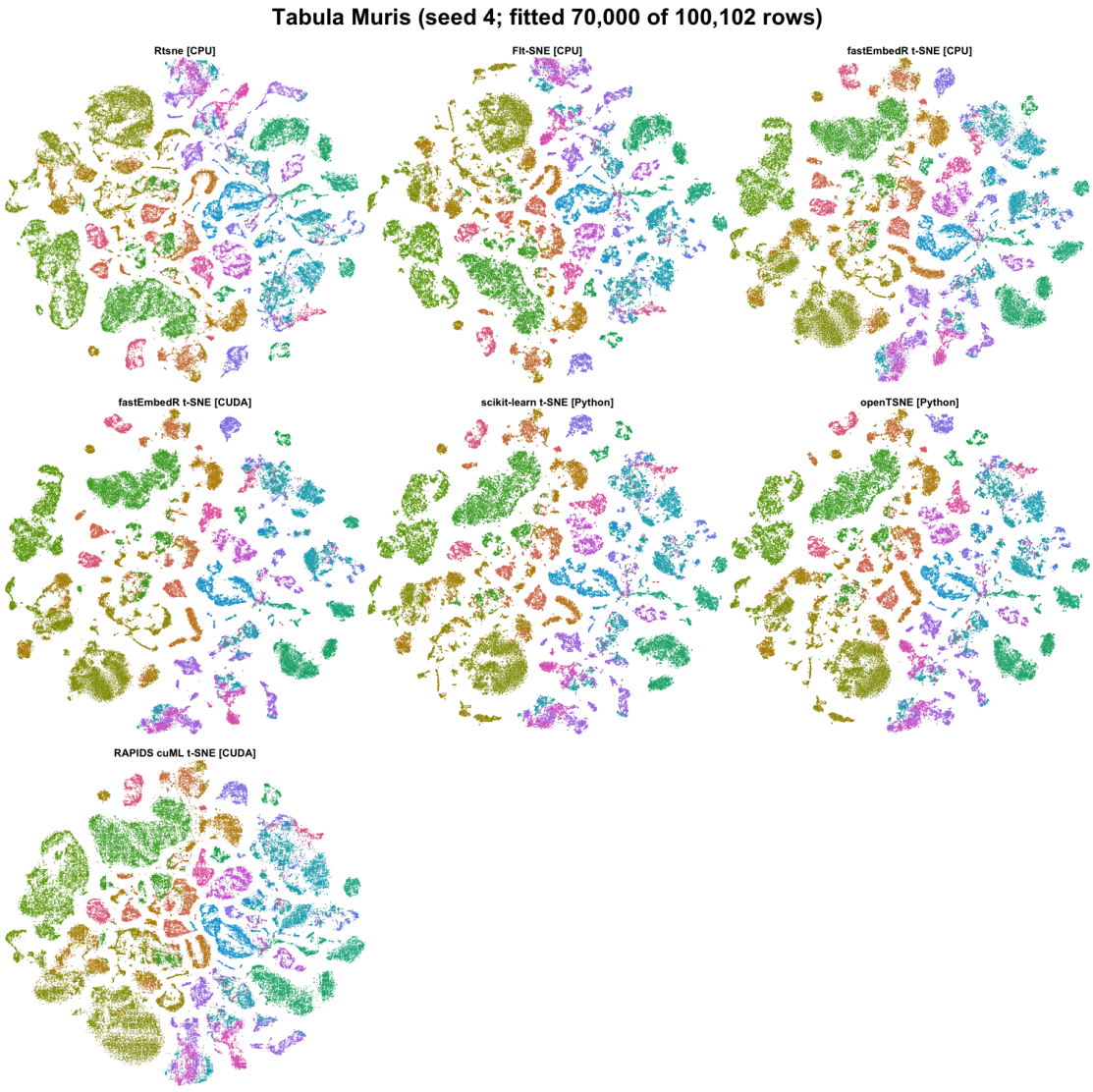

Figure S29: Tabula Muris t-SNE outputs for the tested methods (seed 4; fitted 70,000 of 100,102 rows). Colors encode benchmark labels and were not used for fitting.

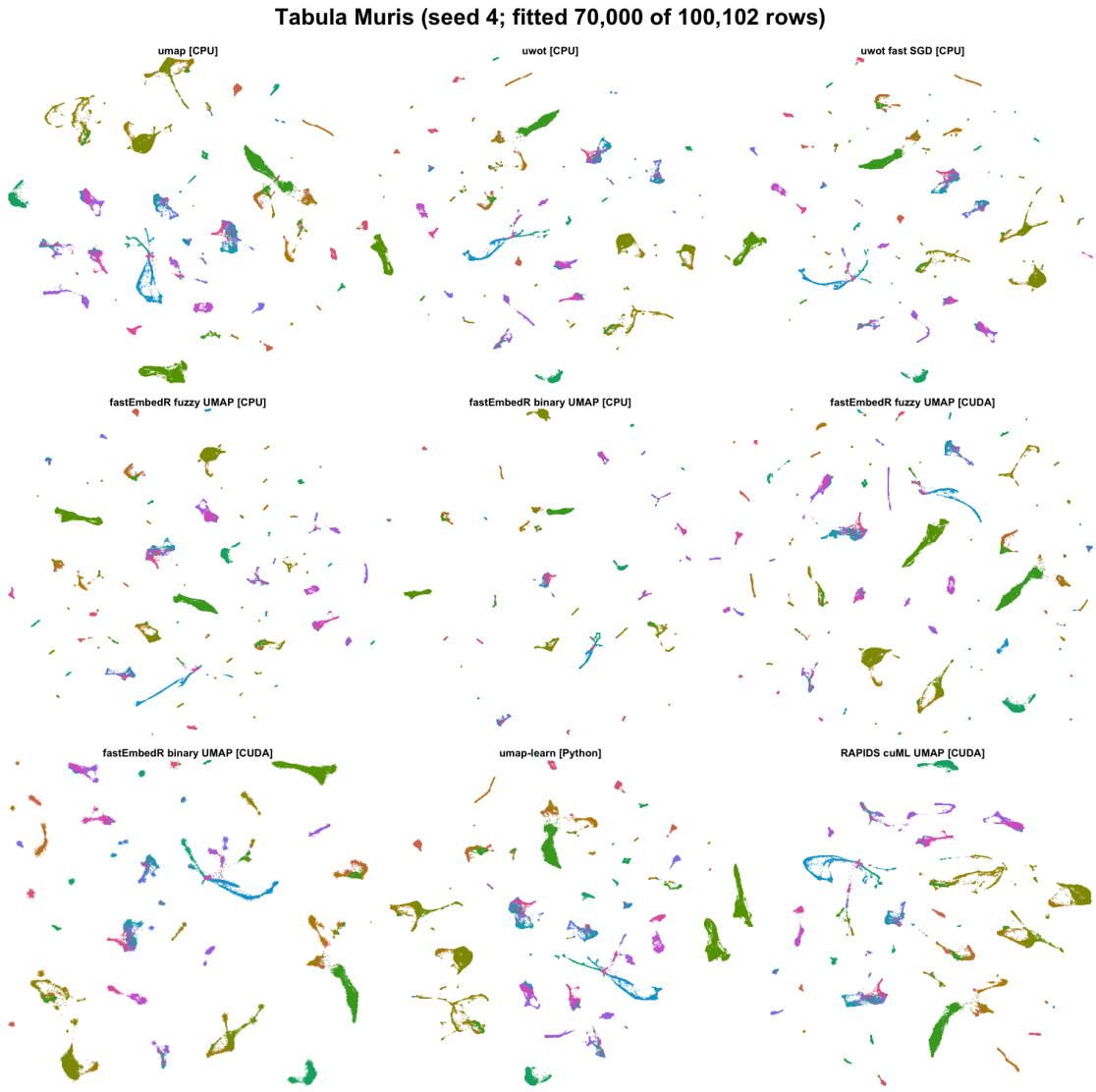

Figure S30: Tabula Muris UMAP outputs for the tested methods (seed 4; fitted 70,000 of 100,102 rows). Colors encode benchmark labels and were not used for fitting.

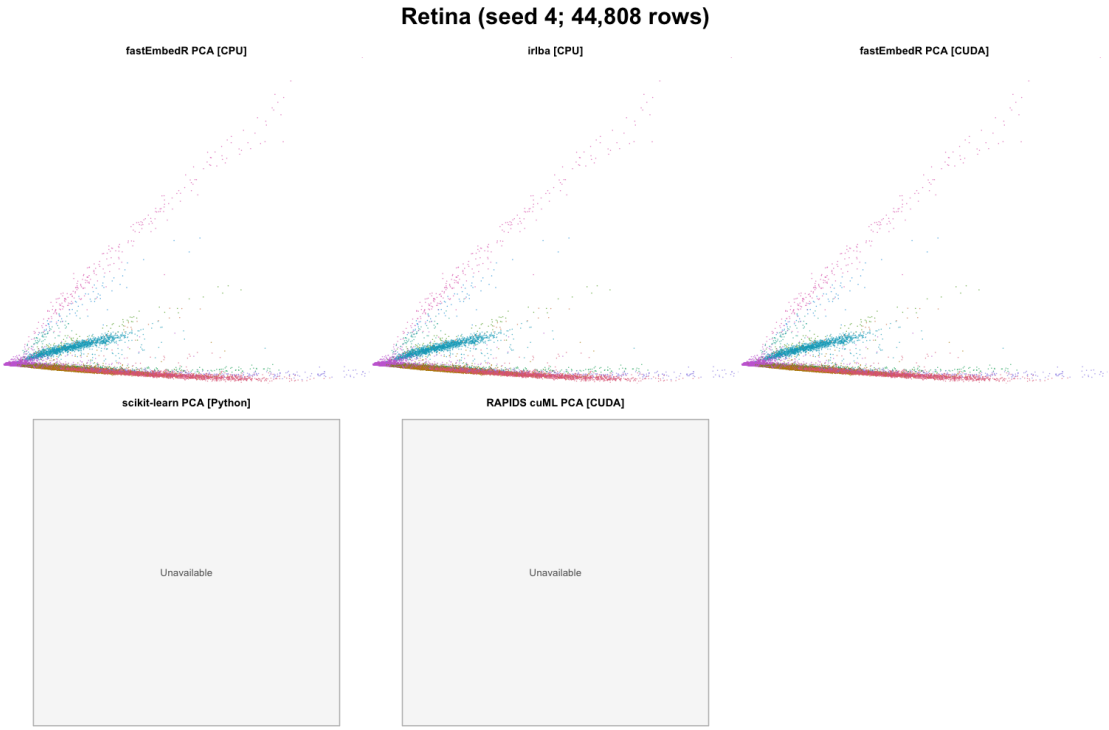

Figure S31: Retina PCA outputs for the tested methods (seed 4; 44,808 rows). Colors encode benchmark labels and were not used for fitting.

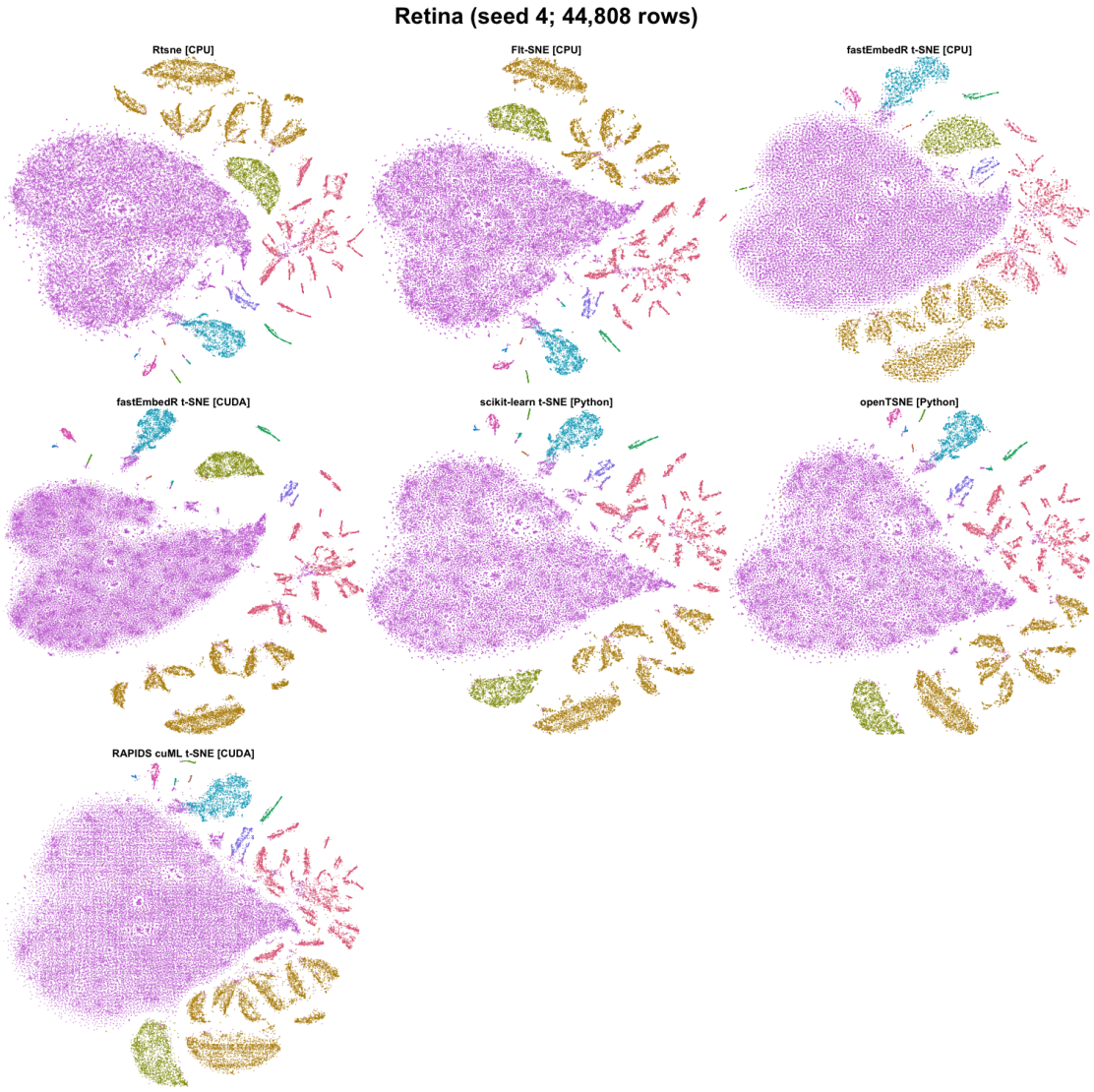

Figure S32: Retina t-SNE outputs for the tested methods (seed 4; 44,808 rows). Colors encode benchmark labels and were not used for fitting.

Figure S33: Retina UMAP outputs for the tested methods (seed 4; 44,808 rows). Colors encode benchmark labels and were not used for fitting.

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URL: https://github.com/tkcaccia/fastEmbedR-extra

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